4

Surface Waves on Passive Surfaces of Finite Extent

4.1 INTRODUCTION

Different kinds of surface waves can propagate along periodic structures. Most familiar is the type attached to a periodic structure situated in a stratified medium. It was shown in reference 72 that as the frequency increases, grating waves will eventually start to propagate along the structure inside the dielectric. If these plane waves are incident upon the dielectric interfaces at angles of incidence larger than the critical angle(s), they will be totally reflected and thus remain inside the stratified medium. The combination of these crisscrossing plane waves looks similar to the well-known surface waves encountered in dielectric slabs; that is, they look like standing waves inside the slab and are evanescent outside. However, these surface waves must in general be sustained by voltages impressed at the terminals of the elements. Furthermore, since no energy escapes from the structure into free space, no energy is delivered to the terminals; that is, the particular impedance term associated with the surface wave will be purely imaginary. Such surface waves are called *forced surface waves*. If the particular impedance term happened to be zero, no impressed terminal voltages are needed to sustain propagation of the surface wave. In that case we have a *free surface wave*. This is the type usually encountered in the classical textbooks [73]. Whether forced or free, this type of surface wave will readily be observed with computer programs based on infinite array theory like, for example, the PMM program. It will be referred to here as Type I.

In contrast, the new type of surface wave introduced in Chapter 1 (Type II) can be encountered whether a dielectric substrate is present or not, but the structure

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must be finite at least in one dimension as illustrated in Fig. 4.1. While the first kind typically shows up only at higher frequencies (in order for grating lobes to exist), the second occurs only at the lower frequencies such as 20–30% below resonance and only if the interelement spacing D_x is $<0.5\lambda$; that is, there must not even be a hint of grating lobes.

The presence of surface waves on finite periodic structures is of concern primarily for two reasons:

1. If used as an FSS to reduce the backscattering, the reradiation from the surface waves can lead to an increase in the total RCS.
2. If used as a phased array, surface waves can lead to a significant variation in scan impedance from element to element, making precise matching difficult (see Figs. 1.3 and 1.5).

In this chapter we shall study the FSS case in more detail and in particular how to control the surface waves. The phased array case will be discussed in Chapter 5.

4.2 MODEL

The finite array will be modeled by a finite number of infinitely long column arrays (also called stick arrays; see Fig. 4.1). This approach has been widely used by several researchers [74–80]. One of them, Usoff, wrote as part of his dissertation [24] the computer program "Scattering from a Periodic Array of Thin Wire Elements" (SPLAT). The excitation can be either in the form of an incident plane wave propagating in the direction $\hat{s} = \hat{x}s_x + \hat{y}s_y + \hat{z}s_z$ (passive case). Or it can be in the form of generators connected to the elements individually (active case). The program can then calculate the bistatic scattered field as well as the scan impedance of each column, to mention just a few features.

Basically the approach is a combination of the plane wave expansion (spectral) and the mutual impedance approach as explained in Chapter 3 and also reference 62.

4.3 THE INFINITE ARRAY CASE

When analyzing the finite array, it is advisable first to review the infinite array case; that is, the finite number of columns shown in Fig. 4.1 becomes infinite as shown earlier in Fig. 1.1. As discussed for example in reference 62 or Chapter 3, infinite arrays are significantly simpler to analyze than finite arrays. In particular, if exposed to an incident plane wave with direction of propagation equal to $\hat{s}$, the element currents are all related to each other by Floquet's Theorem:

$$I_{qm} = I_{00} e^{-j\beta q D_x s_x} e^{-j\beta m D_z s_z}, \tag{4.1}$$

where q equals the column number and m equals the row number (see Fig. 1.1).

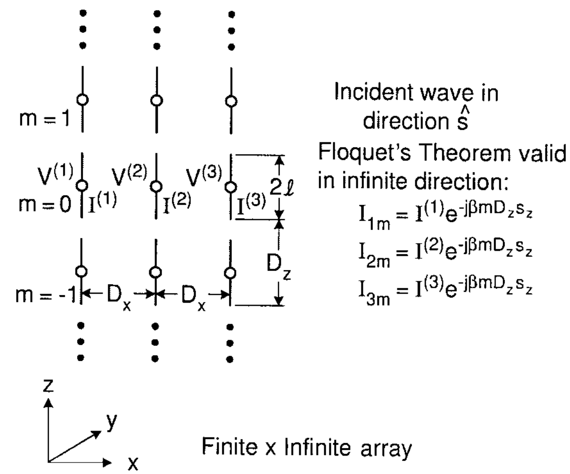


Fig. 4.1 The array considered here consists of a finite number of infinitely long columns (stick arrays) with axial elements. Voltages are induced in all elements by an incident plane wave propagating in the direction $\hat{s} = \hat{x}s_x + \hat{y}s_y + \hat{z}s_z$. We seek the bistatic scattered pattern.

However, when the array has only a finite number of columns as shown in Fig. 4.1, the column currents $I^{(q)}$ differ from column to column in amplitude as well as phase. Only along the infinite Z dimension does Floquet's Theorem apply as indicated in Fig. 4.1.

A typical idealized example of the scan impedance for an infinite array with interelement spacing $D_x/\lambda \sim 0.24$ at $f = 8\,\text{GHz}$ is shown in Fig. 4.2. The total length of the elements is $2l = 1.5$ cm; that is, the array will resonate around 10 GHz. Thus, we observe that the scan impedance Z_A at $f = 8\,\text{GHz}$ will be located in the capacitive part of the complex plane as also seen in Fig. 4.2.

Furthermore, from (4.1) we observe that the phases of the element currents are determined by the direction cosines s_x and s_z. For example, if $s_x = s_z = 0$, all element currents are in phase and the main beam will point in the broadside direction. Similarly, if $s_z = 0$, the beam will scan only in the XY plane or the H plane. And if furthermore $s_x = 0.5$ or $\sqrt{3}/2$, the beam will be scanned to 30° or 60° from boresight, respectively. These typical cases are illustrated in Fig. 4.2 by small inserts adjacent to the various scan impedances $Z_A = R_A + jX_A$. Note in particular that while X_A remains fairly constant with scan angle,[1] R_A increases significantly. In fact, at grazing for $s_x = 1$ or $\eta = 90°$, R_A becomes singular (*Note*: There is no groundplane).

Had we excited the elements passively by an incident plane wave, the entire scan range would be limited to be between $s_x = 0(0°)$ and $s_x = \pm1(\pm 90°)$. However, if we instead excite the array actively by driving the elements by individual voltage generators, the scan range can be extended into the imaginary (or invisible) space.

[1] True only for small D_x/λ.

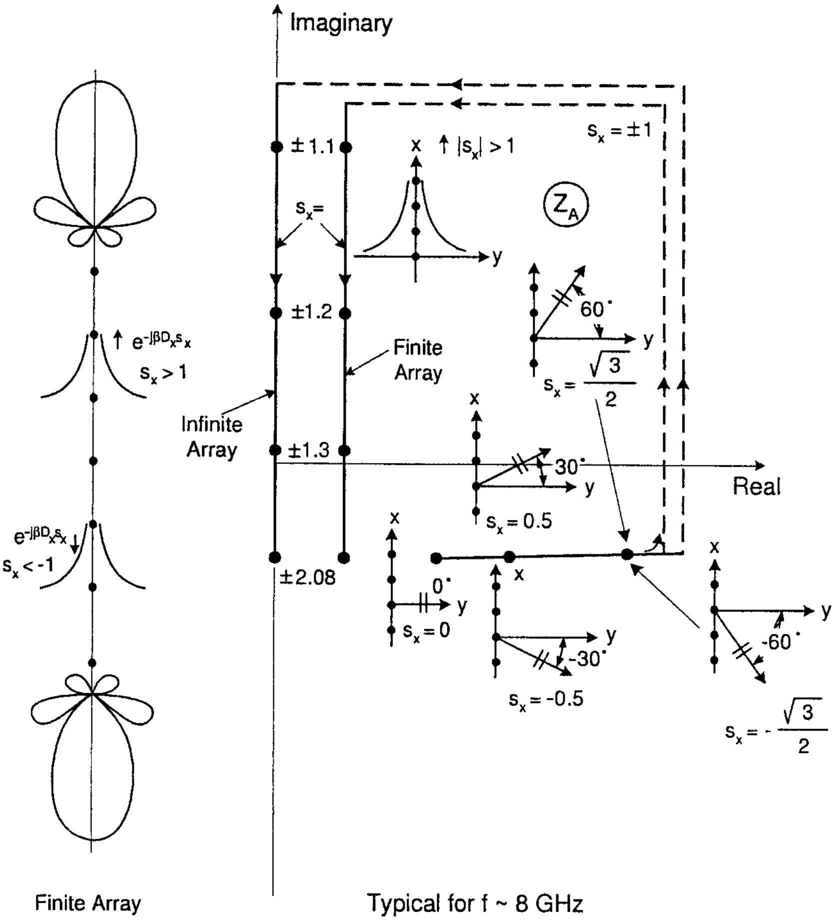


Fig. 4.2 Typical scan impedance for an infinite and a finite array at the fixed frequency 8 GHz (i.e., below resonance) as a function of scan angle. Note: By feeding the elements with actual voltage generators like a phased array, we can scan the "beam" beyond endfire into the imaginary space where $|s_x| > 1$ and only evanescent waves are possible, provided that the interelement spacing D_x is $< 0.5\ \lambda$.

For example, if the interelement spacing $\beta D_x = \pi/2$, then endfire condition is obtained when the phase increment between adjacent elements is equal to 90°. But since the individual generator voltages are controlled entirely by the "operator," he can choose the phase increment to exceed 90° corresponding to $s_x > 1$.

We now recall [61]

$$r_y^2 = 1 - \left(s_x + k\frac{\lambda}{D_x}\right)^2 - \left(s_z + n\frac{\lambda}{D_z}\right)^2. \tag{4.2}$$

We have propagating waves when $0 < r_y^2 < 1$ and evanescent when $r_y^2 < 0$. Thus, the transition between propagating and evanescent waves is determined by $r_y^2 = 0$; that is, from (4.2) we have

$$\left(s_x + k\frac{\lambda}{D_x}\right)^2 + \left(s_z + n\frac{\lambda}{D_z}\right)^2 = 1. \tag{4.3}$$

In the s_x, s_z plane (4.3) depicts unit circles with centers at $k\lambda/D_x, n\lambda/D_z$ as shown in Fig. 4.3.

Every time we are inside one of the unit circles, we are in visible space where we observe propagating modes. When outside, we are in invisible space where the modes are evanescent. For more about the grating lobe diagram, see reference 61.

If $D_x/\lambda = 0.5$ or smaller, we readily see that the unit circles will never touch each other; that is, we will never have a grating lobe, but will have only one propagating wave when particular values of s_x, s_z puts us inside one of the unit circles.

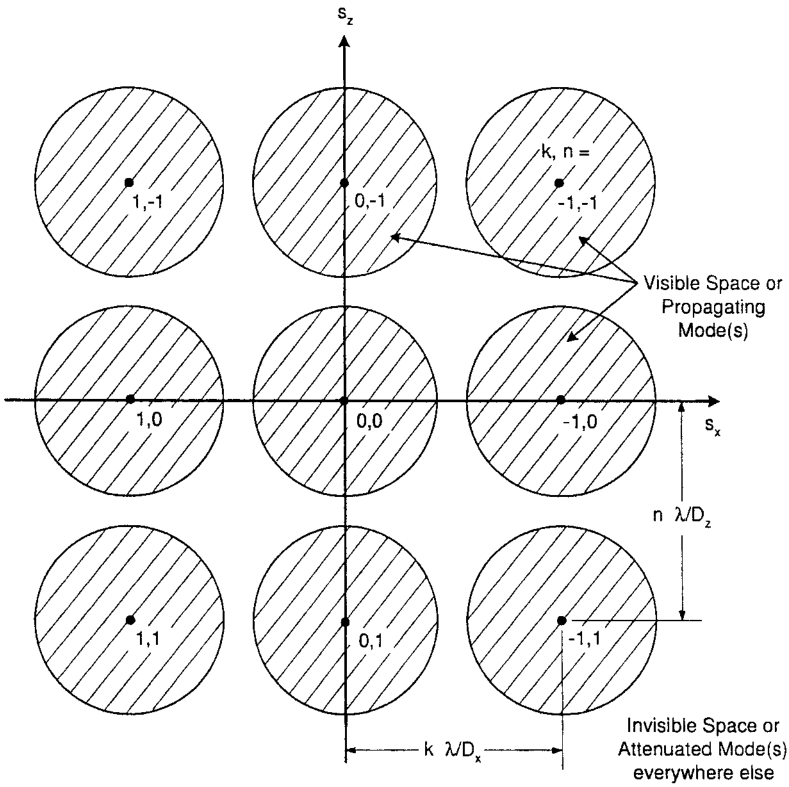


Fig. 4.3 Grating lobe diagram plotted in the (s_x, s_z) plane. If (s_x, s_z) falls inside any of the unit circles, we have propagating mode(s). Otherwise, we have evanescent modes.

The typical example in Fig. 4.2 has $D_x/\lambda = 0.24$ (or $\lambda/D_x = 4.15$). Thus, when s_x increases from $s_x = 1$, the scan impedance Z_A will be located on the imaginary axis and reach the most negative imaginary value for $s_x = \frac{1}{2}\lambda/D_x = 2.08$. Further increase of s_x to $s_x = \lambda/D_x - 1 = 4.15 - 1$ (equivalent to $s_x = -1$; see Fig. 4.3) leads Z_A back up along the imaginary axis the same way it came down as illustrated in Fig. 4.2.

Note in particular that $Z_A = 0$ for $s_x = \pm 1.32$. This case is particularly interesting because this constitutes the condition for a free surface wave as discussed earlier. In other words, a surface wave will propagate even when the impressed voltage is zero. Also note that it is the sum of the individual evanescent waves that constitutes a surface wave that is unique and satisfies the boundary condition at the elements of the array, not the individual modes.

4.4 THE FINITE ARRAY CASE EXCITED BY GENERATORS

We now turn our attention to the finite array shown in Fig. 4.1. We shall at first obtain an *approximation* of this case simply by truncation of the infinite array considered above. We will assume that waves similar to the surface waves on the infinite array can exist on the finite array. However, a significant difference between the two cases is that while no energy is radiated when the array is infinite, radiation will take place in the finite case similar to a finite array with a progressive current radiating in the endfire mode. Typically a free surface wave is encountered at this frequency in the neighborhood of $s_x = \pm 1.32$, where the positive sign produces a pattern radiating upward and the negative sign produces an identical pattern radiating downward (see Fig. 4.2, left). This radiated energy in the finite case must be accounted for by supplying energy to the individual elements using voltage generators. This in turn implies that the scan impedance for $|s_x| > 1$ no longer can be purely imaginary but must contain a real component as indicated in Fig. 4.2 and labeled as “Finite Array.”

To summarize: No energy is radiated from the infinite array, only from the finite. Thus, a test antenna located a reasonably large distance from the array will pick up a signal only from the finite array, not the infinite. Furthermore, from the law of reciprocity we may conclude that a signal from the test antenna cannot produce a surface wave on the infinite array but can readily do so on the finite array.

Because of the lossy component of the scan impedance, the waves on the finite array can be considered as being surface waves for a slightly lossy periodic structure.

4.5 THE ELEMENT CURRENTS ON A FINITE ARRAY EXCITED BY AN INCIDENT WAVE

We now return to our original problem, namely, where a finite array is exposed to an incident field rather than fed by voltage generators at each terminal.

When a plane wave with direction of propagation $\hat{s} = \hat{x}s_x + \hat{y}s_y + \hat{z}s_z$ is incident upon a periodic structure, we would expect to see strong currents of the Floquet type, namely the type given by (4.1). Thus, they all have the same amplitude and a relative phase which matches that of the incident plane wave.

When the array is infinite, these Floquet currents are the *only* ones present. This is a direct consequence of Floquet's Theorem, which basically states that for a periodic structure, the phases of the element currents must match those of the incident field. However, when the structure becomes finite, it is no longer periodic and Floquet's Theorem simply does not apply.

For a *finite* × *infinite* array as shown in Fig. 4.1, we will now show that the current will be:

1. The Floquet type as found on an infinite periodic structure—that is, of the form given by (4.1).
2. Traveling waves with $|s_x| > 1$ propagating in each direction along the array. Each of these will produce an end-fire radiation pattern. They are very similar to surface waves on an infinite periodic structure except that these waves have a loss component due to radiation (see Fig. 4.2).
3. Finally, there will be an end effect associated with the reflection of the three traveling waves at the edges of the array.

While it is easy to accept that the currents on a finite array will be of the Floquet type like they are on an infinite structure, it is more difficult to accept that the two traveling waves exist ONLY on the finite and not on the infinite array. The following discussion will hopefully clarify this problem.

4.6 HOW THE SURFACE WAVES ARE EXCITED ON A FINITE ARRAY

Consider an infinite array exposed to an incident plane wave $\overline{E}^i$ with the direction of propagation being $\hat{s}$, as shown in Fig. 4.4, top. From the basic theory for periodic structures we know that the reradiated (scattered) field consists of plane waves propagating in the directions $\hat{s} = \hat{x}s_x \pm \hat{y}s_y + \hat{z}s_z$ and, eventually, a finite number of grating lobes, as also illustrated in Fig. 4.4, top. Furthermore, there will be an infinite number of evanescent waves that die out quickly as we move away from the array.

Let us next consider a finite array as illustrated in Fig. 4.4, middle. When exposed to the same incident plane wave with the direction of propagation $\hat{s}$, the element currents will again, to a first-order *approximation*, be the Floquet currents as given by (4.1).

It is a simple matter to find the *far* field from these currents as a function of the continuous radiation direction $\hat{r}_c$. We show a typical example in Fig. 4.4, bottom. However, rather than plotting it as a function of radiation angle, it is plotted as a function of r_{cx}; see Fig. 4.4 middle. This makes it more compatible with our choice of variable s_x for the incident wave and the plane wave expansion in general. We obtain mainbeams at $r_{cx} = s_x$ and we also note that the visible

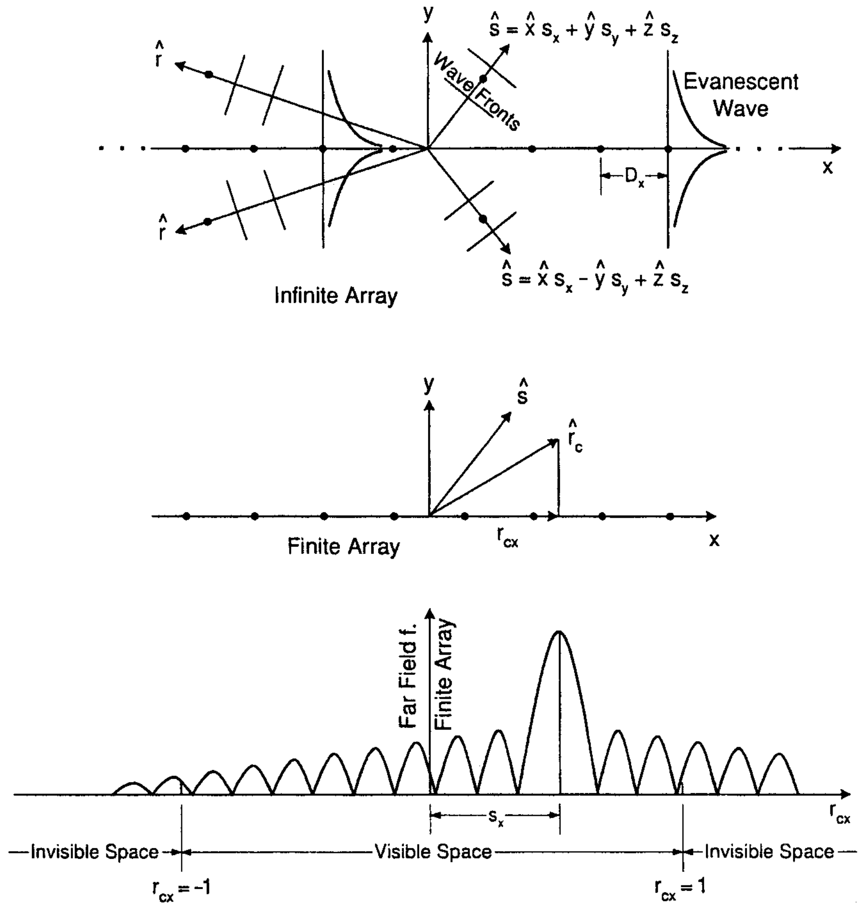


Fig. 4.4 Top: If an infinite array is exposed to an incident plane wave propagating in the directions, $\hat{s}$, it will reradiate propagating plane waves in the directions $\hat{s} = \hat{x}s_x \pm \hat{y}s_y + \hat{z}s_z$ and eventually a finite number of grating waves. In addition, evanescent waves are always present. Middle: A finite array with only Floquet currents will produce a continuous spectrum $\hat{r}_c$ with a mainbeam and sidelobes as shown at the bottom. Note that we use the x component r_{cx} of $\hat{r}$ as our variable. This is consistent with using s_x from the direction $\hat{s}$ of the incident plane wave. Bottom: Far field from Floquet currents as a function of the continuous radiation direction $\hat{r}_c$.

space runs from $r_{cx} = -1$ to $r_{cx} = +1$ as indicated in Fig. 4.4, bottom, while the invisible space is given by $|r_{cx}| > 1$, as shown.

Note also that as we continue into the invisible space for $r_{cx} < -1$ and $r_{cx} > 1$ there is no profound change in the general character of the radiation pattern.

However, the radiation pattern in Fig. 4.4 bottom is only a first-order approximation. To see what really goes on, consider Fig. 4.5. At the very top (Fig. 4.5a) we show an infinite periodic structure being exposed to an incident plane wave with direction of propagation $\hat{s}$. The element currents will be of the Floquet type only as we saw earlier in Fig. 4.4. In Fig. 4.5b we create a finite array by

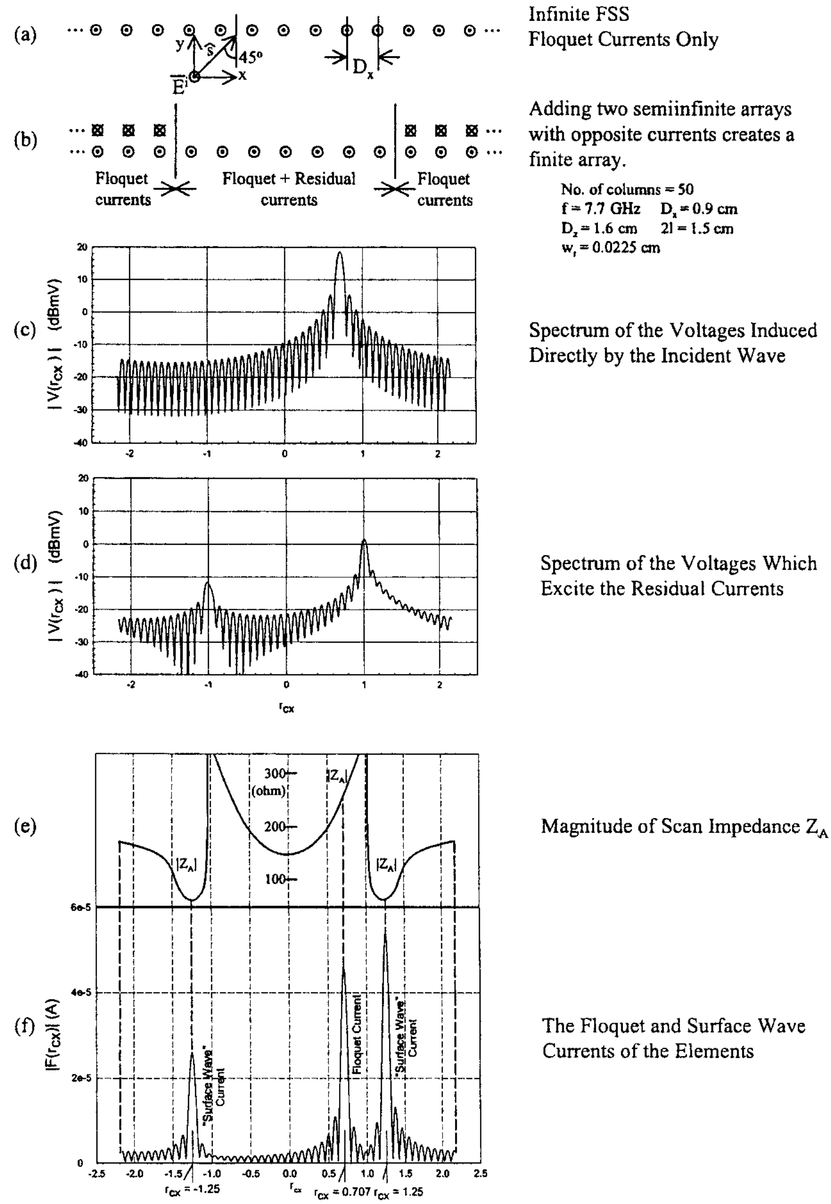


Fig. 4.5 (a) An infinite array exposed to an incident plane wave has only Floquet currents. (b) Adding two semi-infinite arrays with negative Floquet currents creates a finite array with actual currents (Floquet and residual currents). (c) Spectrum of the voltages induced in the finite array by the incident wave. (d) Spectrum of the voltages in the finite array by the two semi-infinite arrays. Note the two peaks in the endfire directions $r_{cx} = \pm 1$. (e) The magnitude of the scan impedance Z_A as a function of r_{cx}. Note the minima at $r_{cx} = \pm 1.25$. (f) The spectrum of the element currents as a function of r_{cx}. Note the surface waves where $|Z_A|$ is minimum is at $r_{cx} = \pm 1.25$, not at $r_{cx} = \pm 1.0$.

superimposing two semi-infinite arrays with Floquet currents that are negatives of the original Floquet currents in the infinite case. Note that the currents on the two semi-infinite arrays are given rigorously by the Floquet currents only [i.e., (4.1)]; however, the currents on the finite array in the middle will, as we shall see, deviate for this simple form.[2] This becomes clear when we realize that the voltages induced in the elements of the finite array have two sources:

1. The incident plane wave
2. The two semi-infinite arrays

The voltage spectrum of the incident wave taken over the finite array is shown in Fig. 4.5c. It contains mainbeams in the forward and specular directions corresponding to the Floquet currents only (see Fig. 4.4).

Similarly, the voltage spectrum from the two semi-infinite arrays is shown in Fig. 4.5d. We observe two smaller beams at $r_{cx} = \pm 1$, that is, in the endfire directions. Thus, the two semi-infinite arrays will try to propagate waves along the array structure. However, as illustrated in Fig. 4.2, many evanescent waves with different $r_{cx}(s_x)$ are capable of propagating. They are distinguished by the magnitude of their scan impedance Z_A depicted in Fig. 4.2 and shown specifically in Fig. 4.5e.

4.7 HOW TO OBTAIN THE ACTUAL CURRENT COMPONENTS

The discussion above served primarily to explain how surface waves are established on a finite periodic surface, namely as the ratio between the voltages induced by the two semi-infinite arrays and the impedances of possible surface waves.

This is not necessarily the way we actually calculate the element currents. In fact this was done by direct calculations of the currents in the finite array in question by using the SPLAT program discussed in Chapter 3. Typical examples have already been presented in Figs. 1.3b and 1.3c. Clearly the currents in Fig. 1.3c are seen to be highly erratic. To find out what current components actually are contained in such a distribution, we simply ran a Fourier analysis and obtained the current spectrum shown in Fig. 4.5f. While the current spike at $r_{cx} = 0.707$ is easy to associate with the Floquet currents obtained for an infinite array exposed to a plane wave incident at 45°, the two other spikes at $r_{cx} = \pm 1.25$ remained somewhat of a mystery until the explanation in Section 4.6 above was introduced.

Finally, the Fourier analysis shows that we in addition to the Floquet and surface waves also obtain a small amount of additional currents at the edges of the array. They are usually associated with reflections of the two surface waves and the Floquet currents at the edges of the array. We have denoted them "end currents."

[2] See also Common Misconceptions, Section 4.19.

It is interesting to observe that the amplitude of one of the surface waves actually is larger than the Floquet current. Inspection of Fig. 1.3c strongly suggests that it may indeed be the case. However, the impact of the surface waves should not be judged by their amplitude only. Both the Floquet and the surface wave currents will radiate (scatter) and we shall see in the next section that the former in general is a far more efficient radiator than the latter.

Note: Sometimes the reader perceives the introduction of the two semi-infinite arrays with negative Floquet currents as an inadequate approximation. As discussed in detail in Section 4.19 (Common Misconceptions), this is not the case. However, even if some inaccuracies were present in our explanation, it really would not matter since the currents in Fig. 4.5f were obtained by direct calculation of the actual currents obtained from the SPLAT program applied to a finite array.

4.8 THE BISTATIC SCATTERED FIELD FROM A FINITE ARRAY

In the previous section we gave a physical explanation of how various current components on finite arrays came about. Furthermore, we used the SPLAT program in conjunction with a Fourier analysis to determine the actual element currents on a finite array. We decomposed these according to their phase velocities and found one strong component at $r_{cx} = s_x = 0.707$ for 45° angle of incidence corresponding to the Floquet currents present on an infinite array while we observed the surface wave currents propagating in opposite directions along the array at $r_{cx} = \pm 1.25$. Finally, there was a component that we associated with reflections from the edges of the finite array (we are not entirely sure about this interpretation at this point). We will refer to them as "end currents."

Furthermore, it is often convenient to introduce the concept of "residual currents," defined as the difference between the actual currents on the finite array and the Floquet currents. In other words, the residual currents are simply here defined as the sum of the two surface waves and the end currents.

These current components will radiate. Their radiation patterns are obtained simply by considering the entire periodic structure as an antenna (the notion that surface waves do not radiate is true only if they are associated with an infinite structure. This is usually the case as presented in most textbooks).

Thus, based on the current components obtained rigorously in the previous section, we shall next present the radiation pattern associated with these components.

First we show in Fig. 4.6 the bistatic scattering pattern of the Floquet currents only when a signal is incident upon an array of 50 columns at 45° angle of incidence. We obtain main beams in the forward as well as the specular directions and note that the sidelobe level looks "clean" as expected (sin x/x function).

This is also an opportune time to remind the reader that the total field in the forward direction is given by the sum of the scattered and the incident field. Since these two components are basically out of phase and have similar magnitudes, the total field in the forward direction is about zero except for some minor sidelobes. Thus, we observe a shadow in the forward direction as we would expect.

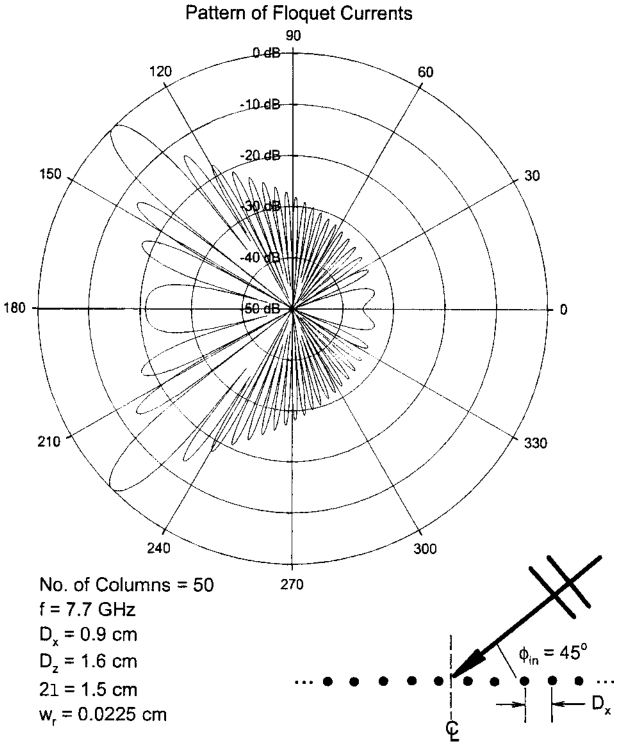


Fig. 4.6 The bistatic scattering pattern for the Floquet currents only. Angle of incidence is 45°.

Furthermore, we show in Fig. 4.7 the bistatic scattering pattern of the two surface waves. We observe that they are identical except for amplitude and direction of propagation. Note that the ratio between the pattern amplitudes of the two surface waves is close to the ratio between the two surface waves' amplitudes given in Fig. 4.5.

However, perhaps the most noteworthy is that in spite of the fact that one of the surface waves has a peak exceeding that of the Floquet currents peak by about 5 dB, the highest value of the scattering pattern is almost 20 dB below the peak of the Floquet pattern. In other words, the radiation efficiency of the surface waves is considerably lower than that of the Floquet currents. Or we may alternatively state that the radiation resistances associated with the surface waves are considerably lower than the one associated with the Floquet mode. Inspection of Fig. 4.5e shows this statement to be correct. This observation will later prove crucial when we try to control the radiation from the surface waves without significantly attenuating the one from the Floquet mode.

Furthermore, we show in Figs. 4.8 and 4.9 the bistatic scattering pattern associated with the end currents only and the residual currents, respectively. Comparing Figs. 4.7 and 4.8 with their sum in Fig. 4.9 indicates that the surface and end currents are basically out of phase. However, see also Section 4.9.

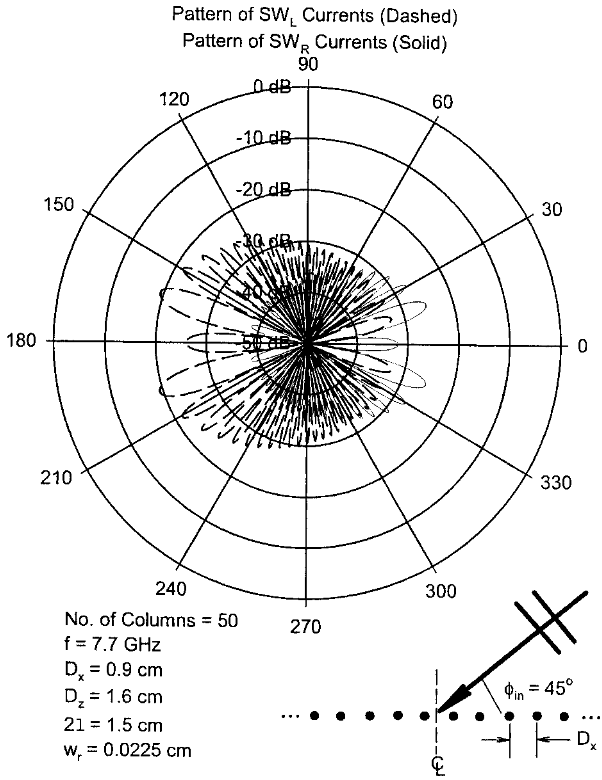


Fig. 4.7 The bistatic scattering pattern for the left- and right-going surface wave. Angle of incidence is 45°.

Finally, we show in Fig. 4.10 (broken line) the bistatic scattering pattern of the actual current on the finite array as obtained from the SPLAT program—that is, the sum of the Floquet currents, the surface waves, and the end currents. This pattern should be compared with the Floquet pattern shown in Fig. 4.6. It has been redrawn in Fig. 4.10 (solid line) for easy comparison. We readily observe that the main beams are unaffected and so is the *location* of the sidelobes. However, the sidelobe level is 5–7 dB higher when we include the radiation from the residual currents.

4.9 PARAMETRIC STUDY

Any periodic structure, whether infinite or finite, will in general change its properties considerably with angle of incidence. Furthermore, if the surface is of finite extent, we would expect its characteristic to depend to a degree on its size. Finally, surface waves are, as seen earlier, only prevalent in a certain frequency band.

Thus, in the following we shall examine some of these features in more detail.

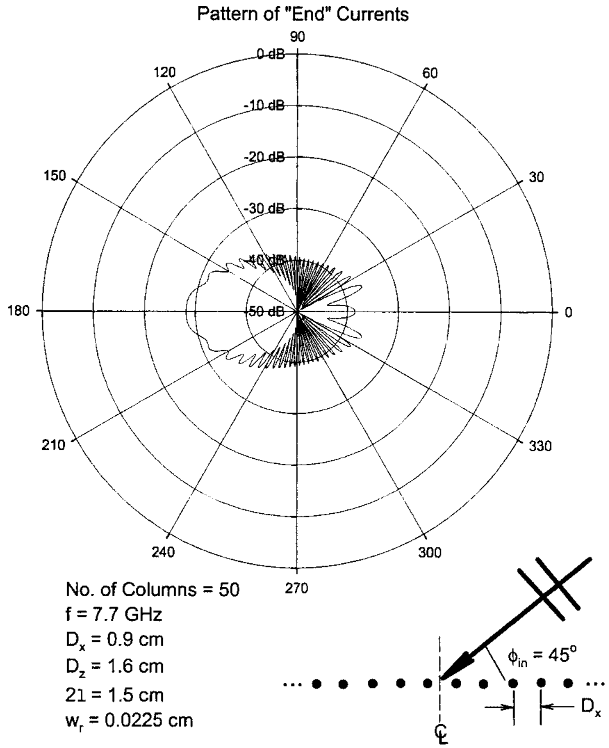


Fig. 4.8 The bistatic scattering pattern of the "end" currents. Angle of incidence is 45°.

4.9.1 Variation of the Angle of Incidence

First, for general orientation, we show in Fig. 4.11 the same case as shown in Fig. 4.10 except that the angle of incidence is now 67.5° (or 22.5° from grazing) instead of 45°. We observe only minor changes in the mainlobes while the location and level of the sidelobes has changed somewhat. However, as we shall see next, it actually takes only a small variation of the angle of incidence to obtain significant changes of the bistatic scattering pattern.

We have already illustrated in Figs. 4.5a and 4.5d how two semi-infinite arrays at each end of the finite array launches surface waves along the finite structure. In Fig. 4.12a we show this scenario in more detail. The semi-infinite arrays at the right of the finite array produces a field essentially pointing to the left. It will launch a left-going surface wave. Similarly, we will also obtain a right-going surface wave. The fields from the two semi-infinite arrays will have a phase difference that depends on the size of the finite array and the angle of incidence as shown in Fig. 4.12b. If we denote the width of the finite array by L and the angle of incidence by θ, the phase difference between the signals from the two semi-infinite arrays will be

$$\Delta = \beta L \sin\theta. \tag{4.4}$$

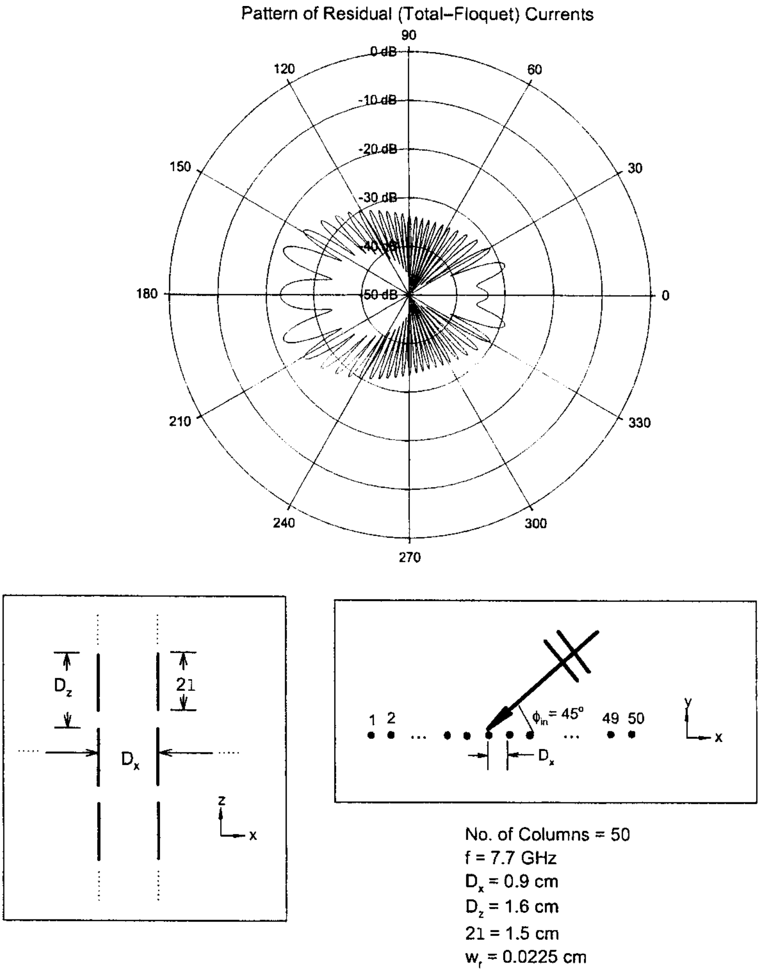


Fig. 4.9 The bistatic scattering pattern of the residual currents (i.e., Total–Floquet's). Angle of incidence is 45°.

Since L in general will be large in terms of wavelength, Δ can be substantial. However, it is of utmost importance that this phase difference is carried directly over in the two surface waves. In other words, the radiation pattern of the two surface waves will go in and out of phase as the angle of incidence θ changes. Let us consider an example.

Example 1 We assume a typical array consisting of 50 columns has an interelement spacing $D_x = 0.9\,\text{cm}$. Then the total width of the finite array is $L = 50 \times 0.9 = 45\,\text{cm}$. Thus, for an angle of incidence $\theta = 45°$ and frequency $f = 7.7\,\text{GHz}$ we obtain a phase difference between the left- and right-going surface wave equal to

$$\Delta = \beta L \sin\theta = 16.35\pi \ \text{(rad)}.$$

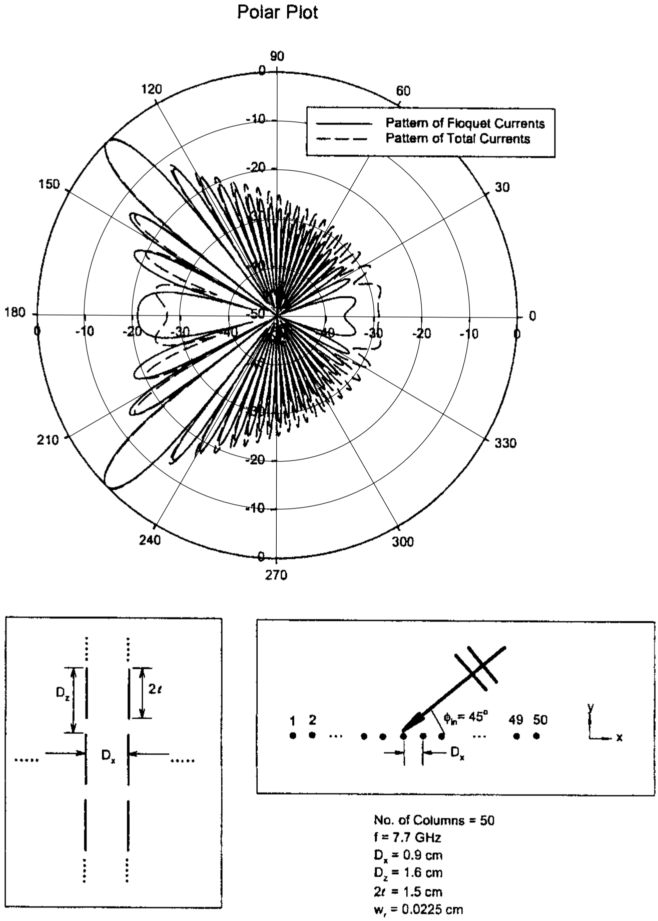


Fig. 4.10 The bistatic scattering pattern of the Total as well as the Floquet currents. Angle of incidence is 45°. Note the increase in sidelobe level when the residual currents are added.

We now reduce the angle of incidence by $\Delta\theta$ such that the phase difference between the two surface waves is reduced by π rad, that is,

$$\sin(\theta - \Delta\theta) = \frac{(16.35 - 1)\pi}{\beta L} = 0.665.$$

In other words, $\Delta\theta = 3.4^\circ$ for a 180° phase change between the two surface waves.

We will illustrate this result in Fig. 4.13. We here show the calculated scattering pattern for an array of columns and vary the angle of incidence from 45.5° to 41.0° in steps of 0.2°. We then selected the bistatic scattering pattern with the strongest residual pattern, namely Fig. 4.13a for $\theta = 45.2^\circ$ and the weakest residual pattern as shown in Fig. 4.13c for $\theta - \Delta\theta = 41.8^\circ$. Thus to obtain a

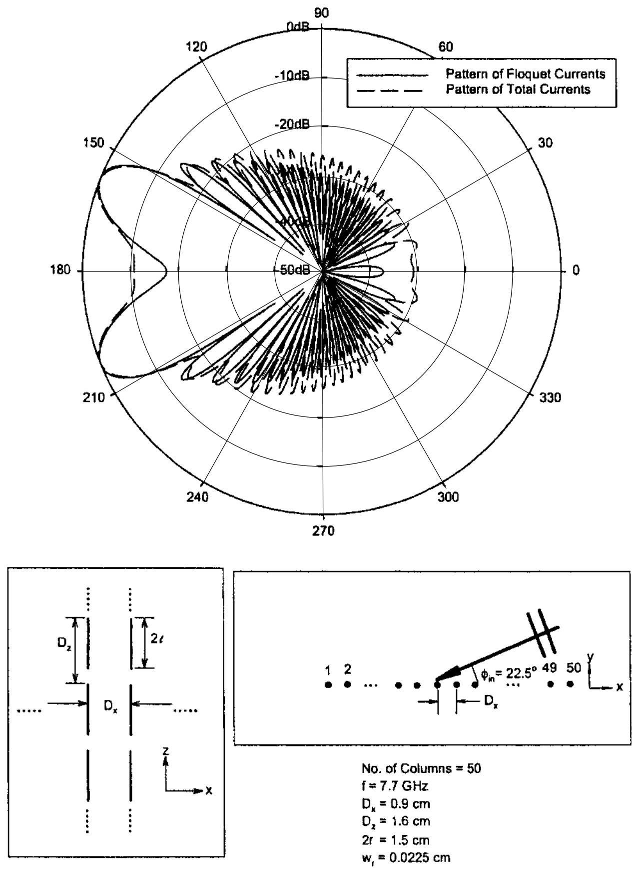


Fig. 4.11 The bistatic scattering pattern of the Total as well as the Floquet currents. Angle of incidence is 67.5°.

phase reversal of the two surface waves (including the end pattern), the angle of incidence should change by $\Delta\theta = 45.2 - 41.8 = 3.4^\circ$, in complete agreement with the estimated value calculated above. As an extra check, we show a scattering pattern in between as illustrated in Fig. 4.13b.

4.9.2 Variation of the Array Size

Inspection of (4.4) above showed that the phase difference between the two semi-infinite arrays could be changed by variation of the angle of incident θ. Alternatively, a change in phase difference can also be obtained by variation of the width L of the finite array.

More specifically, let us change the original width L of the finite array to $L + \Delta L$ such that the phase difference Δ is changed by π; that is, by modification of (4.4) we find

$$\Delta + \pi = \beta(L + \Delta L)\sin\theta. \tag{4.5}$$

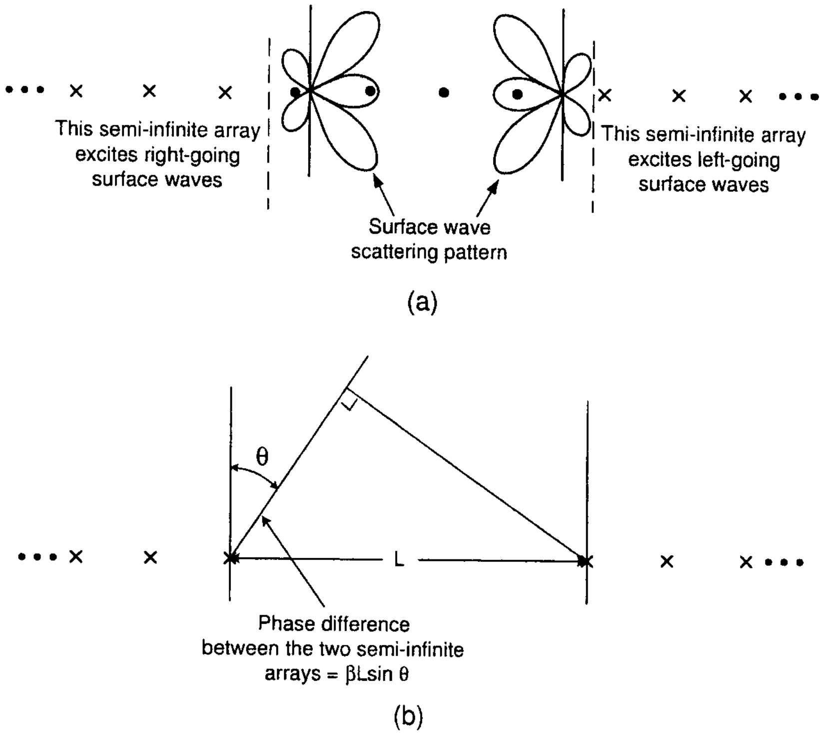


Fig. 4.12 (a) The two semi-infinite arrays excite right- and left-going surface waves as indicated in the figure. (b) Since the width L of the finite array usually is large in terms of wavelength, the phase difference between the two semi-infinite arrays will in general change strongly with angle of incidence θ. This phase difference is carried directly over into the two surface waves.

From (4.4) and (4.5) we obtain

$$\Delta L = \frac{\lambda}{2\sin\theta} \tag{4.6}$$

or the change ΔN in number of elements is

$$\Delta N \sim \frac{\Delta L}{D_x} = \frac{\lambda}{2D_x \sin\theta} \tag{4.7}$$

Example 2 Typically for $D_x = 0.9\,\text{cm}$, $\theta = 45°$, and $f = 7.7\,\text{GHz}$, we find from (4.7)

$$\Delta N \sim 3.1 \text{ column.} \tag{4.8}$$

Note: ΔN is independent of the width L of the array.

This estimate has been checked out by calculating the bistatic scattering pattern for arrays where the number of columns vary from 50 to 55 in increments of one. In Fig. 4.14a we show the case where the residual pattern is close to a maximum for $N = 51$ columns, and in Fig. 4.14b we show the case where it is close to a

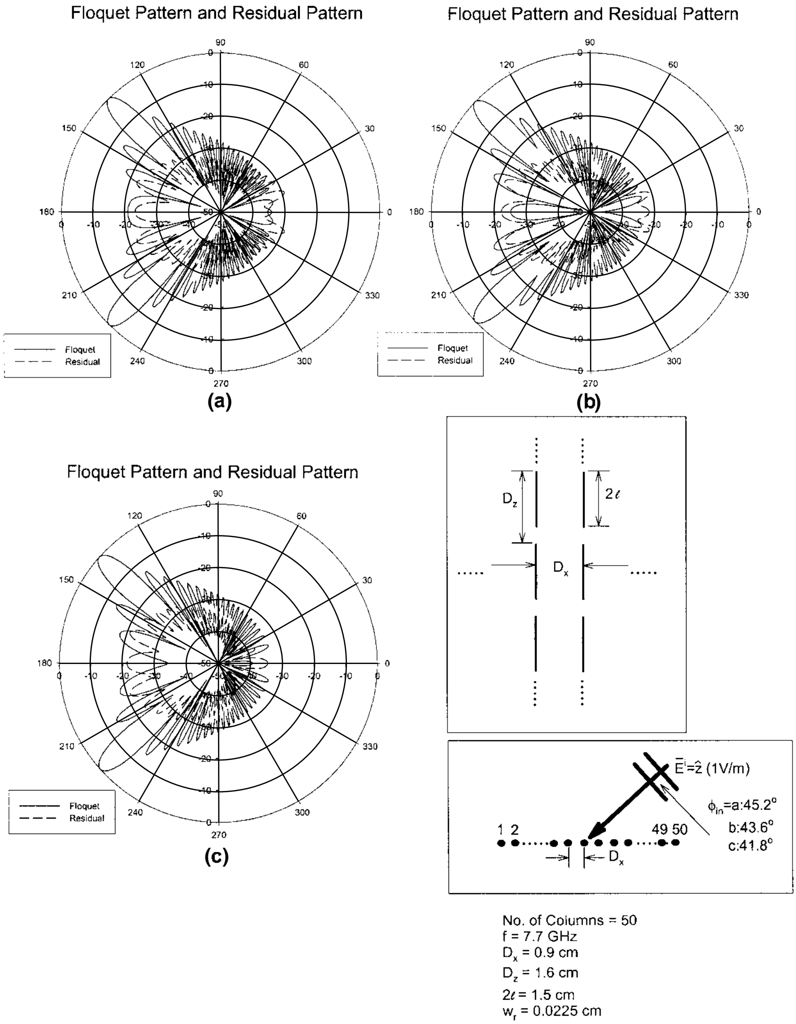
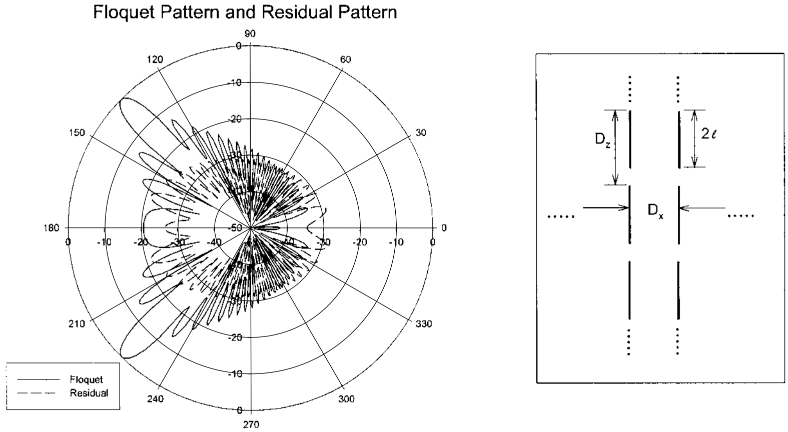


Fig. 4.13 The bistatic scattered fields for an array of 50 columns at $f = 7.7$ GHz. Floquet currents only (full line) and residual currents only (broken line). Angle of incidence: (a) 45.2°. Fields from the residual currents are maximum. (b) 43.6°. Fields from residual currents are medium. (c) 41.8°. Field from residual currents are minimum. The amplitude of the residual fields depends on the phase difference between the two semi-infinite arrays shown in Fig. 4.12.

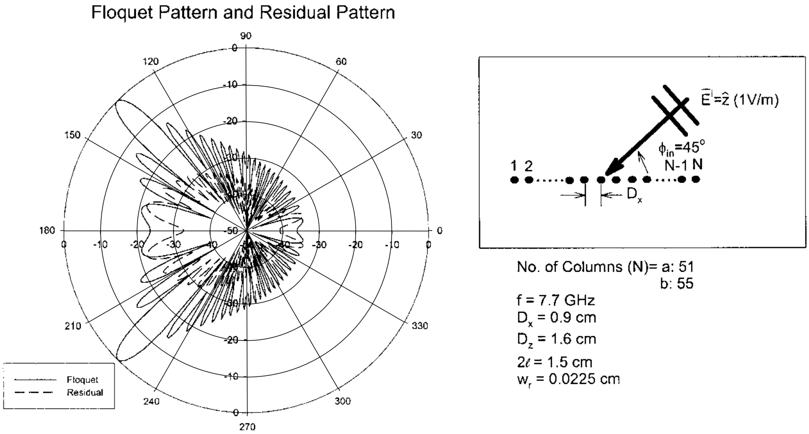
minimum for 55 columns. Thus, $\Delta N = 55 - 51 = 4$ columns, which is in "fair" agreement with the estimate above. *Note*: The presence of the end current pattern makes "exact" comparison of the residual pattern somewhat blurred.

4.9.3 Variation of Frequency

All the examples presented so far have been at the frequency $f = 7.7$ GHZ. The effect of changing the frequency will be investigated next. The size of the array



(a)



(b)

Fig. 4.14 The bistatic scattered field from arrays with (a) 51 columns and (b) 55 columns. Full line denotes field from Floquet currents only, and broken line denotes field from the residual currents only. Angle of incidence is 45°. Note that the fields from the residual currents are maximum for 51 columns and minimum for 55 columns.

is kept fixed at 25 columns, and a plane wave is incident at 67.5° from normal. We will calculate and plot the backscattered field in the complex plane as shown in Fig. 4.15. At the top of that figure we show the frequency range 2–6.2 GHz and at the bottom the range 6.3–12 GHz. We note that the low frequencies from 2 to 6.2 GHz rotate in the complex plane in a very regular and leisurely fashion as one would expect. However, in the frequency range 6.3–8.3 GHz shown at the bottom we observe a backscattered field that not only is significantly larger but also rotates much faster in the complex plane. Finally, from 8.4 to 12 GHz the backscattered field still has a large amplitude but has settled down and is rotating

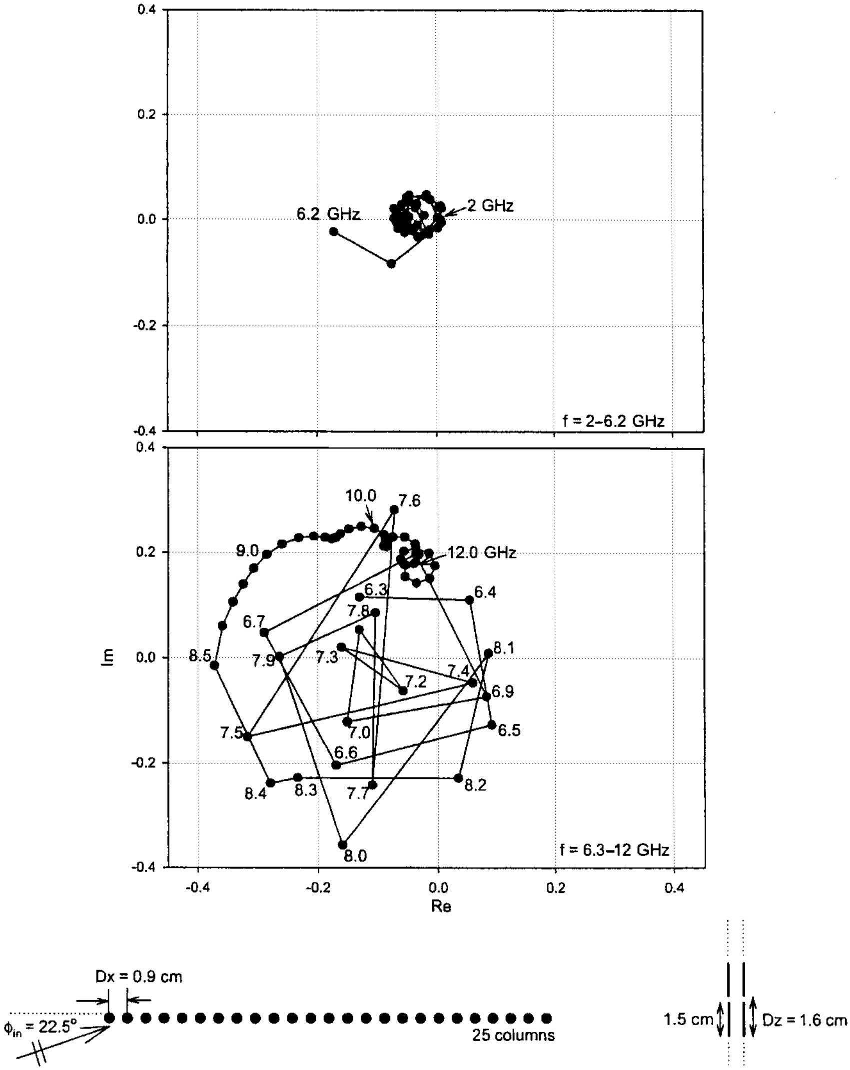


Fig. 4.15 The backscattered field plotted in the complex plane as a function of frequency. Angle of incidence is 67.5°. Number of columns is 25. Top: Frequency range 2–6 GHz. Bottom: Frequency range 6.3–12 GHz, where the surface waves are prevalent from 6.3 to 8.4 GHz. Called a backscattering diagram.

much slower. A closer investigation shows that the fast rotating range from 6.3 to 8.3 GHz is the frequency range where surface waves exist. We simply in that range observe column currents as shown earlier, for example, in Fig. 1.3 (more examples will be given later; see, for example, Figs. 4.17, 4.18, and 5.9). From 8.5 to 12 GHz the surface waves have essentially died out. There are two reasons for the higher values of the backscattered field in this frequency range.

1. We are closer to the resonant frequency (~10 GHz), resulting in a stronger column current and thus backscattered field.
2. The sidelobe level of the backscattered field is higher because we are getting closer to the onset of the first grating lobe (~18 GHz).

But how do we explain that surface waves are present only in a limited frequency range, namely ~6.3–8.3 GHz? This is explained in Fig. 4.16 in a qualitative way. It shows the scan impedance Z_A plotted earlier in Fig. 4.2 but here plotted at three frequencies, namely 6, 7.7, and 10 GHz. Furthermore, these scan impedance curves are shown a bit more realistic by the fact that they for end-fire condition do no go to infinity but just to a large value depending on how large the array actually is. This point is easy to see by application of the mutual impedance concept. It simply tells that the magnitude $|Z_A|$ of the scan impedance can never exceed $\sum_{q=-Q}^{Q} |Z_{0,q}|$, where $Z_{0,q}$ is the mutual impedance between the reference element in column 0 and all the elements in column q (see Chapter 3 for details). Since $|Z_{0,q}|$ and Q are bounded, so is the finite sum $|Z_A|$. As already shown in Fig. 4.2 and repeated in Fig. 4.16 for easy comparison, we

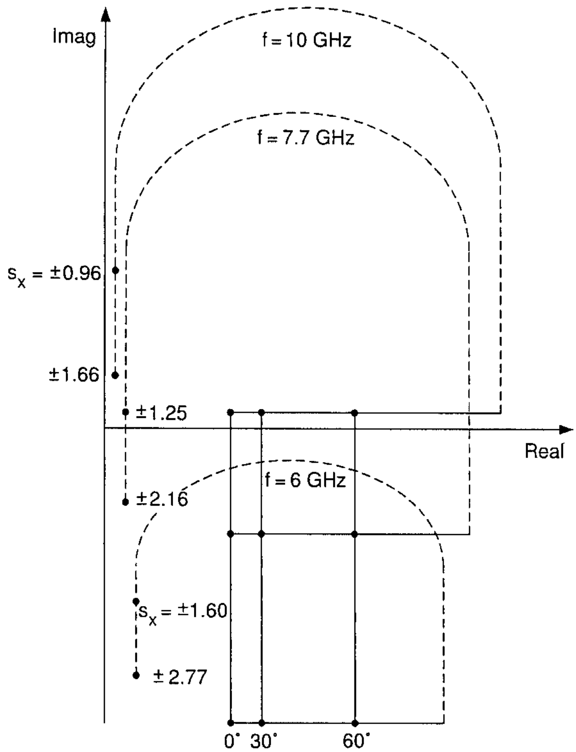


Fig. 4.16 Typical complex scan impedance Z_A for a finite array at the following frequencies: 10 GHz, no surface waves since lowest point for $s_x = \pm 1.66$ is too far from (0,0); 7.7 GHz, surface wave possible for $s_x \sim \pm 1.25$; 6.0 GHz, no surface wave since too far from (0,0).

see that at $f = 7.7\,\text{GHz}$ Z_A gets close to the origin for $s_x = \pm 1.25$; that is, a free surface wave can exist. However, at 10 GHz we are closer to resonance and Z_A will therefore in real space move closer to the real axis in the complex plane. Note that this upward motion is also taking place in imaginary space such that even for $s_x = \pm 1.66$ where Z_A turns around and moves upward along the imaginary axis, Z_A will not get close to the origin; that is, no free surface waves can exist at 10 GHz.

Similarly, at $f = 6\,\text{GHz}$ we drop too far down in the capacitive region for Z_A to get close to the origin; that is, no surface waves can exist here either. In other words: A finite array can only support surface waves in a limited frequency range as illustrated by the actual calculated curve in Fig. 4.15 and only for $D_x < \lambda/2$

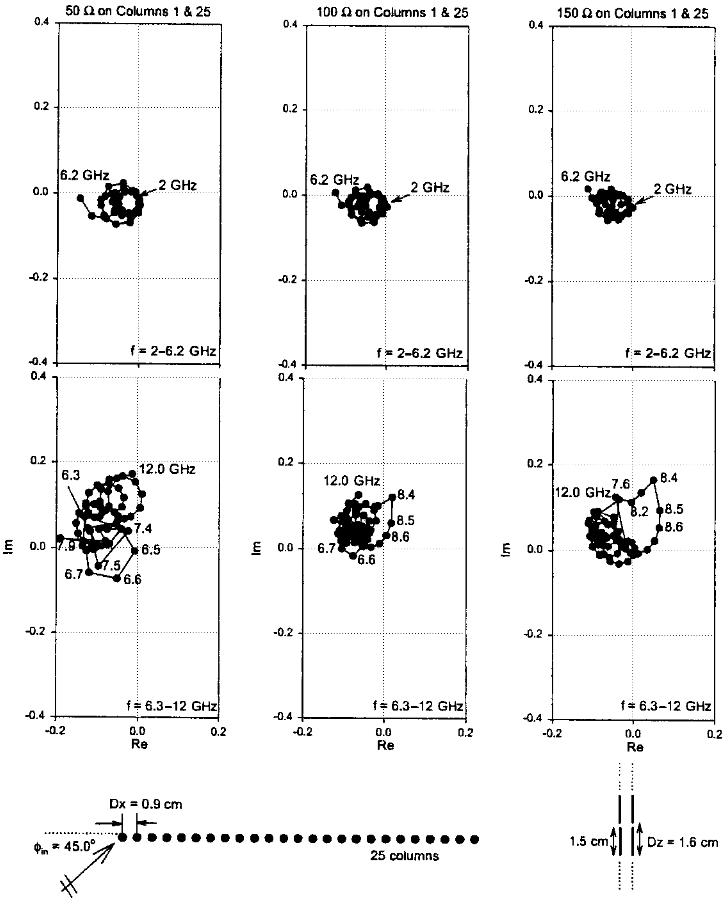


Fig. 4.17 The backscattered field (backscattering diagram) from an array of 25 columns plotted in the complex plane. Frequency range 2–6.2 GHz (top) and 6.3–12.0 GHz (bottom). Outer columns (i.e., numbers 1 and 25) are loaded with (left): 50 ohms; (middle) 100 ohms; and (right) 150 ohms.

(otherwise grating lobes will add a resistive component to the scan impedance and prevent the scan impedance from getting close to the origin). This presentation also shows that infinitely long wires cannot produce surface waves as discussed here. They have no resonance.

The curves shown in Fig. 4.16 are only typical. In reality the values of Z_A will vary from column to column and sometimes even have a small negative real part indicating that energy is absorbed. But the fundamental explanation stands. Note also that the cutoff frequencies will depend to some extent on the size of the array.

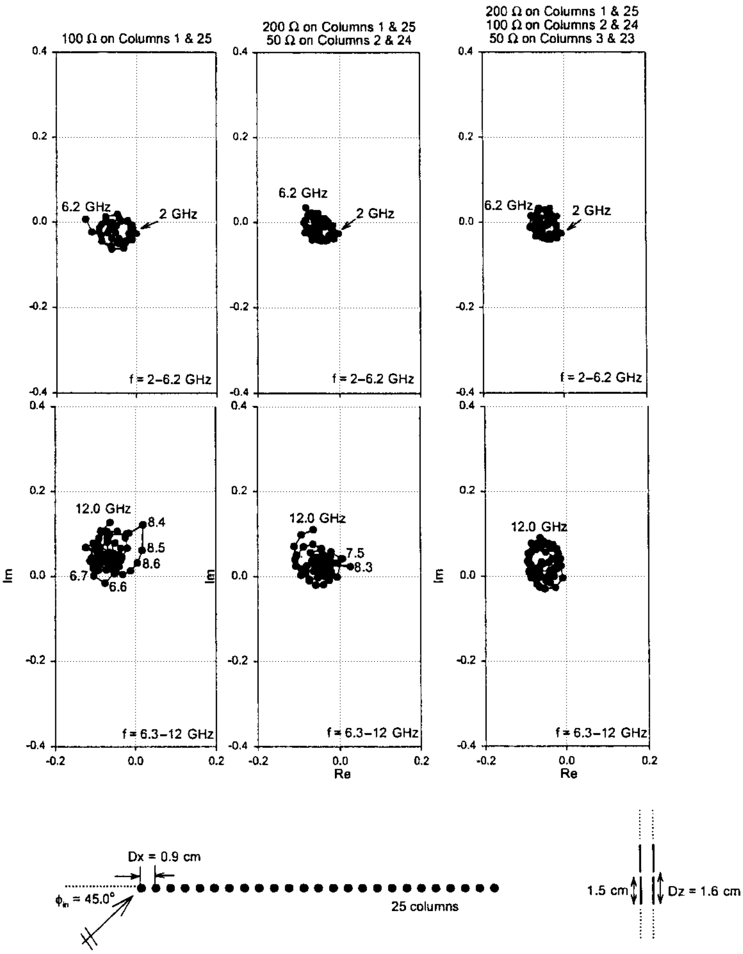


Fig. 4.18 The backscattered field (backscattering diagram) from an array of 25 columns plotted in the complex plane at the frequencies 2–6.2 GHz (top) and 6.3–12.0 GHz (bottom). (left) Columns 1 and 25 loaded with 100 ohms; (middle) columns 1 and 25 loaded with 200 ohms and columns 2 and 24 loaded with 50 ohms; (right) columns 1 and 25 loaded with 200 ohms, columns 2 and 24 loaded with 100 ohms, and columns 3 and 23 loaded with 50 ohms.

4.10 HOW TO CONTROL SURFACE WAVES

We saw earlier that surface waves radiate and can lead to a significant increase in the backscattered field. It is therefore of great interest to investigate ways to control them. We recall from Fig. 4.5d as well as Fig. 4.12 that the surface waves basically were driven by two semi-infinite arrays located on each side of the finite array. Thus, if we could somehow introduce a "barrier" between the two semi-infinite arrays and the finite array, we would expect a weaker excitation of the surface waves in the finite array. One such possible practical arrangement could consist of a finite number of columns between the semi-infinite and finite arrays where the column currents had been reduced by insertion of load resistors in each element. Such an arrangement could also serve as absorbers of the two surface waves as well as the Floquet waves incident upon the edges of the finite array.

Thus, in Fig. 4.17 we show three cases where the elements in the edge columns (one at each edge) of the array shown earlier in Fig. 4.15 have been loaded with 50, 100, and 150 ohms, respectively. We observe only minor changes at the low frequencies 2–6.2 GHz where surface waves do not exist. However, a significant reduction is obtained at the higher frequencies 6.3–12.0 GHz, where surface waves are prevalent from 6.3 to 8.5 GHz (see Fig. 4.15). Note further that the greatest reduction is obtained when the load resistors are 100 ohms (see Fig. 4.17, middle).

Furthermore, we show in Fig. 4.18 three cases where one, two, and three columns at each end of the finite array have been loaded with various resistors as indicated at the top of the figure. First, to the left we show the single column case, namely the optimum case shown earlier in Fig. 4.17, middle. Next follows the two- and three-column cases as shown in Fig. 4.18, middle and right, respectively. As we would expect, a steady improvement is observed as we increase the number of columns.

4.11 FINE TUNING THE LOAD RESISTORS AT A SINGLE FREQUENCY

In the last section we demonstrated how resistive loading of one or more columns located at the edges of the finite array could lead to a significant reduction of the surface waves. We considered the entire frequency range of interest, but our choice of a resistive profile was based on a combination of intuition and experience.

In this section we shall demonstrate a more systematic and also more precise approach. In general, it is performed only at a single frequency. However, experience has shown that the performance at the rest of the frequency band will in general be superior as well.

More specifically we show in Fig. 4.19a the column currents for an array comprised of 25 columns at $f = 7.8$ GHz. The angle of incidence is 45° as shown in the insert below in the figure. There are no resistive loads at all, and this case therefore serves as our baseline. Next we show in Fig. 4.19b the same array but where the three outer columns are loaded with 200, 100, and 50 ohms,

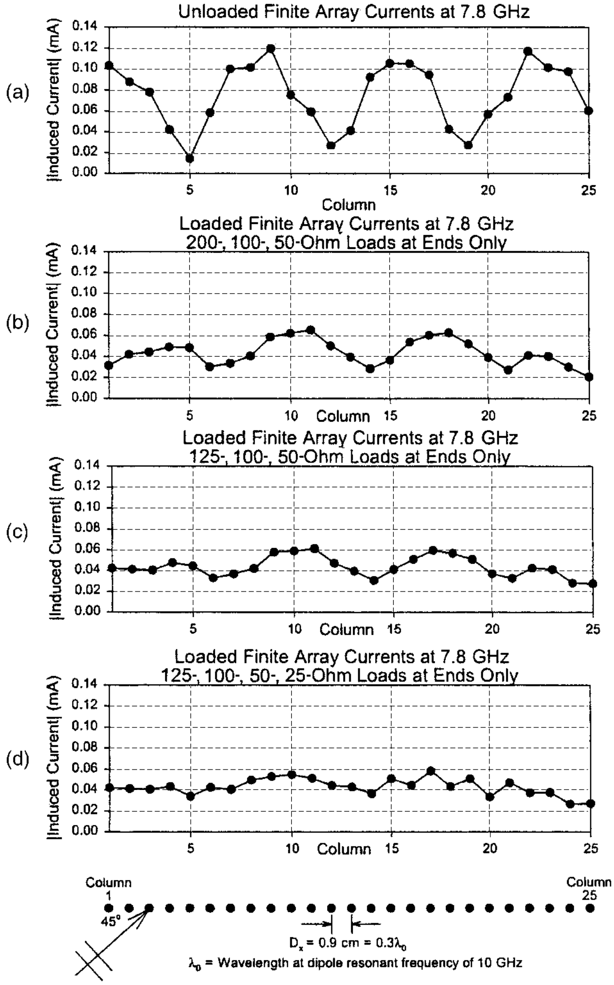


Fig. 4.19 The column currents for an array of 25 columns under various load conditions: (a) No loads (baseline case); (b) columns 1 and 25, 200 ohms, columns 2 and 24, 100 ohms, columns 3 and 23, 50 ohms; (c) columns 1 and 25, 125 ohms, columns 2 and 24, 100 ohms, columns 3 and 23, 50 ohms, (d) columns 1 and 25, 125 ohms, columns 2 and 24, 100 ohms, columns 3 and 23, 50 ohms, columns 4 and 22, 25 ohms.

respectively (actually the same case as shown earlier in Fig. 4.18, right). We observe a substantial reduction of the surface waves as compared with our baseline case in Fig. 4.19a. However, there is still a strong presence of surface waves as indicated by the rather strong variation of the column currents. Obviously the goal is to obtain as even column currents as possible. Close inspection of the currents in Fig. 4.19b reveals that the currents in the two outer columns (nos. 1 and 25) both are somewhat low. Using lower load resistors in these two columns should raise the current. Thus, we show in Fig. 4.19c the case where the load resistors in the outer columns has been reduced from 200 ohms to 125 ohms. We observe that the currents in the three outer loaded columns are indeed fairly constant. However, the equalization of the currents in the rest of the array still needs improvement.

This observation suggests that more columns should be resistively loaded. Thus, we show in Fig. 4.19d a case where the four outermost columns have been loaded with 125, 100, 50, and 25 ohms, respectively. We observe a substantial improvement as compared to all the other cases in Fig. 4.19.

We finally show the backscattering diagram in Fig. 4.20 for the case in Fig. 4.19d. If we compare this case to the case in Fig. 4.18, right, we may at first glance be somewhat disappointed. However, a closer look reveals that a reduction has taken place at the lower frequency band 2–6.2 GHz. Furthermore, the higher values observed at 8.5–12 GHz is as we saw earlier not related to the surface waves. In fact they are "sidelobes" associated with the impending grating lobe about to pop up at $f \sim 18$ GHz. The higher values observed in Fig. 4.20 as compared to Fig. 4.18, right, is probably because the aperture is more tapered in the latter case than in the former (125 ohms compared to 200 ohms). In fact the

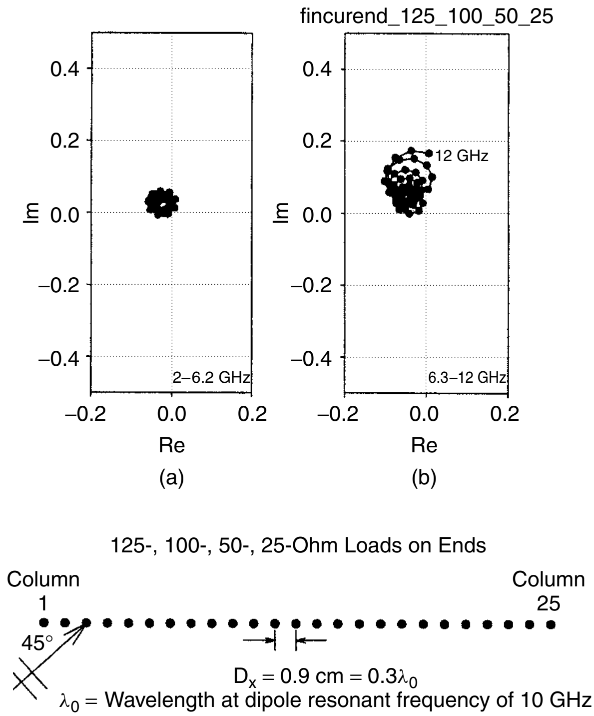


Fig. 4.20 The backscattered field from an array of 25 columns plotted in the complex plane (backscattering diagram). Frequency range: (a) 2–6.2 GHz and (b) 6.3–12 GHz. Load conditions like Fig. 4.19d: columns 1 and 25, 125 ohms; columns 2 and 24, 100 ohms; columns 3 and 23, 50 ohms; columns 4 and 22, 25 ohms.

frequency range from 6.3 to 8.5 GHz where surface waves typically reside looks "cleaner" and more deterministic than the case shown in Fig. 4.18, right.

4.12 VARIATION WITH ANGLE OF INCIDENCE

Periodic surfaces will often be exposed to signals where incidence varies from broadside to high angles. It is well known that the performance of any periodic structure may change considerable in such a scenario (see, for example, reference 81). Thus, it becomes important to check the cases considered above at other angles of incidence than merely 45°.

As an example, let us consider the last case shown in Fig. 4.19d. We show this case again in Fig. 4.21a to facilitate the comparison to follow. Similarly, we show in Fig. 4.21b the same array configuration, but this time when the angle of incidence is equal to 67.5° (or 22.5° from grazing). Note that while the 45°

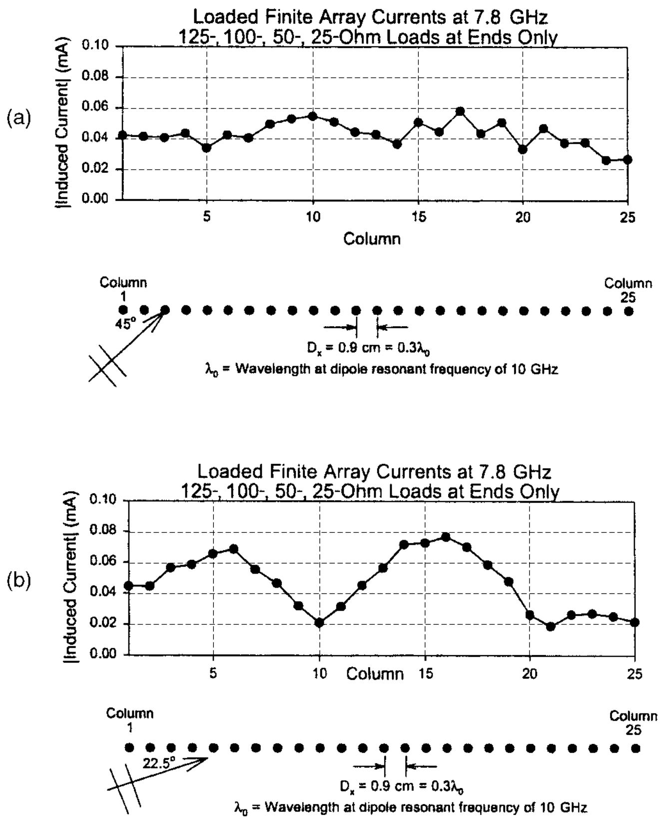


Fig. 4.21 The column currents for an array of 25 columns. Load conditions like Fig. 4.19d: columns 1 and 25, 125 ohms; columns 2 and 24, 100 ohms; columns 3 and 23, 50 ohms; columns 4 and 22, 25 ohms. (a) Angle of incidence is 45°, (b) Angle of incidence is 67.5°.

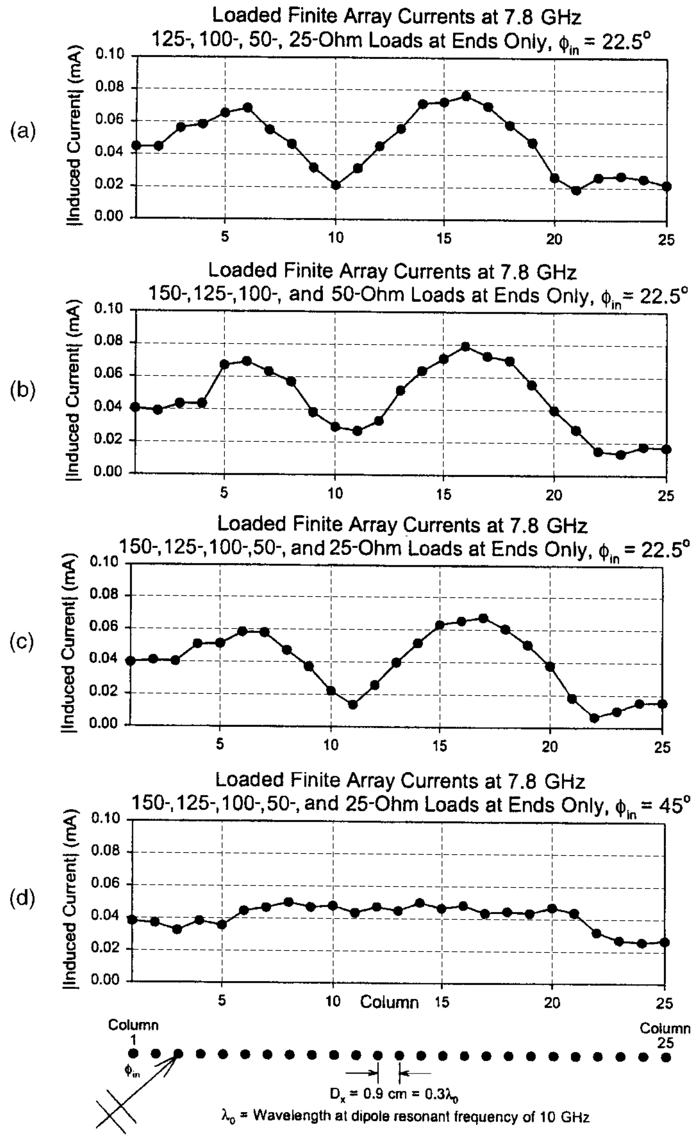


Fig. 4.22 The column currents for an array of 25 columns with various loads. (a) Angle of incidence is 67.5°. Columns 1 and 25, 125 ohms; columns 2 and 24, 100 ohms; columns 3 and 23, 50 ohms; columns 4 and 22, 25 ohms (like Fig. 4.21b). (b) Angle of incidence is 67.5°. Columns 1 and 25, 150 ohms; columns 2 and 24, 125 ohms; columns 3 and 23, 100 ohms; columns 4 and 22, 50 ohms. (c) Angle of incidence is 67.5°. Columns 1 and 25, 150 ohms; columns 2 and 24, 125 ohms; columns 3 and 23, 100 ohms; columns 4 and 22, 50 ohms; columns 5 and 21, 25 ohms. (d) Angle of incidence is 45°. Columns 1 and 25, 150 ohms; columns 2 and 24, 125 ohms; columns 3 and 23, 100 ohms; columns 4 and 22, 50 ohms; columns 5 and 21, 25 ohms.

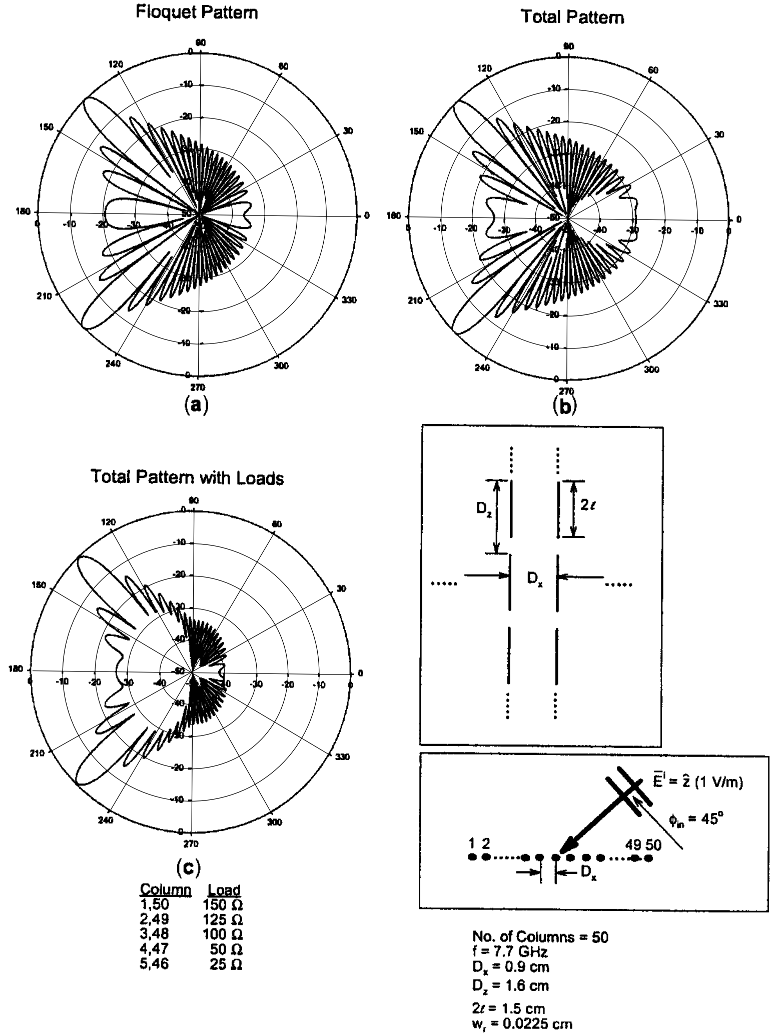


Fig. 4.23 The bistatic scattered field for an array of 50 columns for angle of incidence equal to 45°. (a) Assuming truncated Floquet currents only. No resistive loading. (b) Using the correct total column current as being equal to the sum of the Floquet, the left- and right-going surface waves and the end currents. No resistive loading. (c) Using the correct total column currents as being the sum of the Floquet, the left- and right-going surface waves, and the end currents. Resistive loading as indicated under (c) and same as in Figs. 4.22c and 4.22d. Note strong reduction of the backscatter lobes.

case basically exhibits ~4 humps (most clearly observed in the untreated array shown in Fig. 4.19a, we observe only ~2 humps in the 67.5° case shown in Fig. 4.21b. The reason for this difference is simply that while the phase velocities of the surface waves in the two cases are basically the same, the Floquet currents will move along the structure with a higher phase velocity in the 67.5°

case (i.e., closer to the velocity of the surface waves), resulting in a longer interference wavelength.

A most important consequence of this observation is simply that we need more columns to "sample" the array when the interference wavelength is longer (i.e., at high angle of incidence). As our baseline we show the four-column case already shown in Fig. 4.21b again in Fig. 4.22a in order to facilitate the comparison to follow. Next in Fig. 4.22b we show the case where we have increased all four column loads in an attempt to reduce the column currents. We observe in Fig. 4.22b that this is indeed the case, but there is a jump in column current between columns 4 and 5 and the currents in the rest of the array have not changed significantly.

However, if we load columns 5 and 20 with 25 ohms each, we obtain a notable improvement as shown in Fig. 4.22c.

We finally test the five-column case at angle of incidence equal to 45° as shown in Fig. 4.22d. We observe that the five-column case is quite a bit better than the four-column case shown earlier in Fig. 4.21a.

If the array is reasonably large, the energy lost in the end columns will be relatively minor in average. Thus, in that case the rule seems to be that adding a few extra resistively loaded columns is better than having too few. However, cost must also be considered.

4.13 THE BISTATIC SCATTERED FIELD

In the previous section we obtained an optimum load profile by simply observing the individual column currents in a finite array as illustrated, for example, in Fig. 4.22. We should, however, not lose sight of the fact that if the structure is to be used as a radome, it is the bistatic scattering pattern that must be the final proof of concept. Thus, although we are fairly sure that the optimum load profile arrived at in the above manner also constitutes the optimum profile for the bistatic scattered field, it should nevertheless be checked out. Thus, we show in Fig. 4.23 the bistatic scattering pattern for a *finite* $\times$ *infinite* array with $N = 50$ columns and angle of incidence equal to 45°. More specifically we show in Fig. 4.23a the bistatic scattered field from the finite array by assuming that the column currents are simply of the Floquet types truncated to the size of the array. Similarly we showed in Fig. 4.23b the bistatic scattered field by assuming the correct column current comprised of the sum of the Floquet currents, the two surface waves as well as the end currents (denoted Total Pattern). We observed a significant increase in the sidelobe level for the total pattern as compared to the Floquet pattern, in agreement with earlier observations; see, for example, Figs. 4.10 and 4.11. Finally we show in Fig. 4.23c the same finite array as shown in Figs. 4.23a and 4.23b but where we use the same load profile as used in Figs. 4.22c and 4.22d. We observe that the sidelobe radiation level is significantly lower for the loaded total pattern in Fig. 4.23c than for the unloaded one in Fig. 4.23b (We should certainly expect that!). However, it is even lower than the truncated Floquet pattern in Fig. 4.23a. It seems fairly safe to state that this is related to the fact that the aperture distribution in Fig. 4.23c is tapered as compared to Fig. 4.23a, which is uniform.

We finally show in Fig. 4.24 the same three cases as shown in Fig. 4.23 but where the angle of incidence is 67.5° rather than 45°. Again as in Fig. 4.23 we observe a significantly lower scattering level for the loaded case in Fig. 4.24c as compared to the total unloaded case in Fig. 4.24b (~10 dB) but even up to several decibels below the scattering level in the Floquet case, Fig. 4.24a).

We finally remind the reader that the reduction level is not unique but depends on the size of the array as well as the angle of incidence (see Fig. 4.14 and Fig. 4.13, respectively).

4.14 PREVIOUS WORK

It is considered courteous and scholarly to present other researchers' contributions ahead of your own. However, if your subject is somewhat obscure and perhaps even controversial, this sequence often leads to a clouded relationship between new and old contributions. In fact, reference often end up being merely a litany of titles that "must be there" in order not to offend anyone. My own experience along those lines has often been that although I might have been referenced, it was doubtful whether my contribution was ever read or at least understood. Simply put, the references are simply not meaningful to the reader until after he has read most of the paper.

Thus, in an attempt to alleviate this problem I will review other contributions at this point. Please understand that this sequence in no way is an attempt to put other researchers' work in the background.

It would probably be safe to assume that the first researchers to consider a finite array of passive elements were the inventors of the Yagi–Uda array [82]. They excited these elements by a single active dipole being somewhat different than excitation by an incident plane wave. Besides, the surface wave concept apparently did not occur to them, nor did they actually need it to explain the radiation mechanism. That point of view appears first to have been introduced by Ehrenspeck [83] and later using a more theoretical approach by Mailloux [84]. It is interesting to note that a desirable radiation pattern is not obtained when we have a strong (i.e., free) surface wave, for example, at $s_x \sim 1.25$ as illustrated in Fig. 4.7. We observe that this leads to sidelobes larger than the mainlobe. Rather, the optimum antenna pattern for a Yagi–Uda array is obtained somewhere in the range $1.25 < s_x < 1.0$.

Recall that $s_x = 1.0$ corresponds to the endfire condition, leading to a gain significantly below that of a Yagi–Uda array tuned to maximum gain (it is close to the Hansen–Woodyard condition).

Further significant contribution to the theory of surface waves on Yagi–Uda arrays were given by Richmond and Garbacz [85] as well as by Damon [86].

Surface waves on finite periodic structures of the type discussed in this chapter seem not to have been treated in great detail until recently. That is not to say that

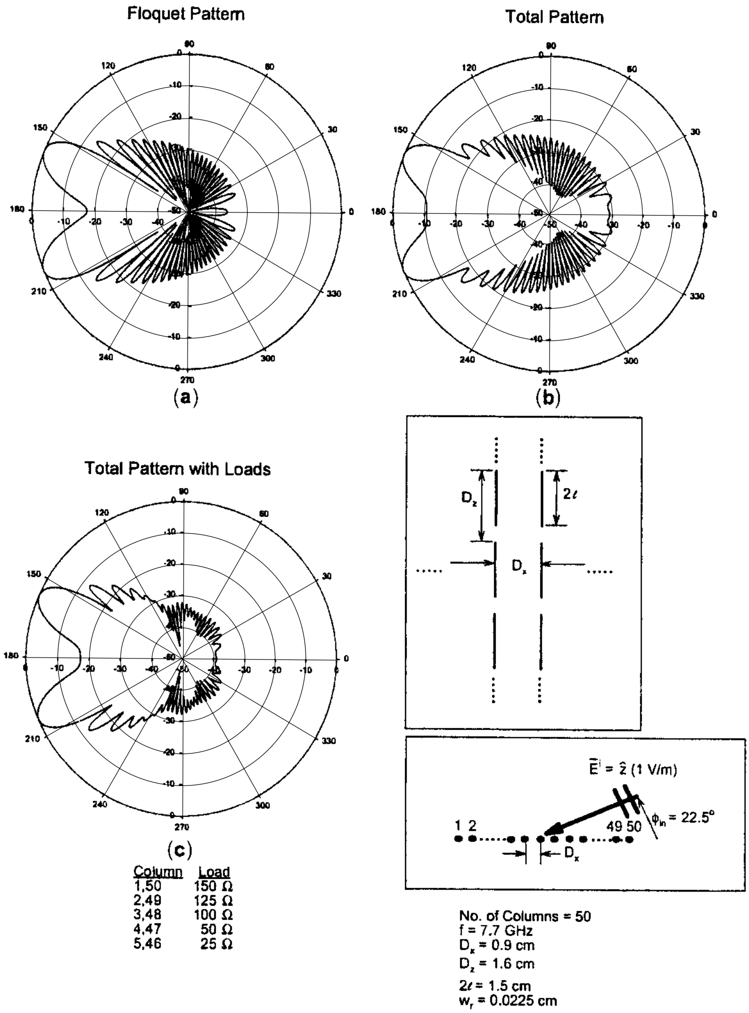


Fig. 4.24 The bistatic scattered field for an array of 50 columns for angle of incidence equal to 67.5° (similar to Fig. 4.23 except for angle of incidence). (a) Assuming truncated Floquet current only. No resistive loading. (b) Using the correct total column currents as being the sum of the Floquet, the left- and right-going surface wave, and the end currents. No resistive loading. (c) Using the correct total column currents as being the sum of the Floquet, the left- and right-going surface wave, and the end currents. Resistive loading as indicated under (c) and same as in Figs. 4.22c and 4.22d. Note strong reduction of the backscatter lobes.

surface waves on periodic structures never were suspected. In fact every time a curious (and often undesirable) effect on a periodic structure was encountered that could not be readily explained, it was often blamed on a mysterious "big bad surface wave." Many have eluded to such phenomena and reported them [3–5, 8–14], while probably even more have suspected them but not reported them. See also Section 1.5.3.

At any rate, the characterization of the surface waves on a finite FSS structure by their actual propagation constant appears not to have occurred until recently in papers by Munk et al. [87] and Janning and Munk [88], as well as in the dissertation by Janning [77] and the thesis by Pryor [76].

It is instructive to speculate on the reasons for this scant attention. Some of them appear to be related to the following:

1. Instead of finite wire sections, infinite long wires (or narrow strips) were used; that is, instead of a double periodic surface, only a single periodic one was used. These cannot support surface waves of the type discussed here (see Section 4.9.3).
2. The frequency band investigated was not sufficiently below the resonance of the structure (20–30%). For further discussion see Section 4.9.3.
3. The interelement spacings D_x were not less than $\lambda/2$; that is, grating lobes could occur as we move into imaginary space. As discussed in Section 4.9.3, that automatically rules out free surface waves.
4. The column currents were simply assumed to be truncated Floquet currents only. In other words the existence of any surface waves were simply ruled out from the onset. See also discussion in Section 4.8.
5. Finally, it should be pointed out that surface waves are naturally attached to planar surfaces. If these are curved, the surface waves will be naturally attenuated by shedding energy as they move along the curved surface. Since many practical applications of periodic structures are naturally curved (for example, such as bandpass radomes and subreflectors), it is easy to understand that the presence of surface waves simply went unnoticed.

4.15 ON SCATTERING FROM FACETED RADOMES

So far we have considered mostly scattering from planar *finite* × *infinite* arrays. They play an important role in helping us to understand the nature of surface waves on finite structures. They are, however, used rarely in practice. They may be curved, or planar sections may be put together in a faceted fashion such as shown, for example, in Fig. 4.25. Curving a surface leads to a natural attenuation of the surface waves by shedding energy. This case will not be treated here (if anyone out there has something to say about that subject, I would appreciate hearing about it).

The shape shown in Fig. 4.25 is characteristic for ship board applications. The incident signal is basically arriving in the horizontal plane. If the radome is opaque, the signal will be reflected upward or downward and thereby lead to a reduced RCS. When the radome is transparent, the incident signal will proceed to the antenna and be received. The RCS will in that case depend essentially on the antenna only. For a discussion of this subject, see Chapter 2. Also see Pryor [76].

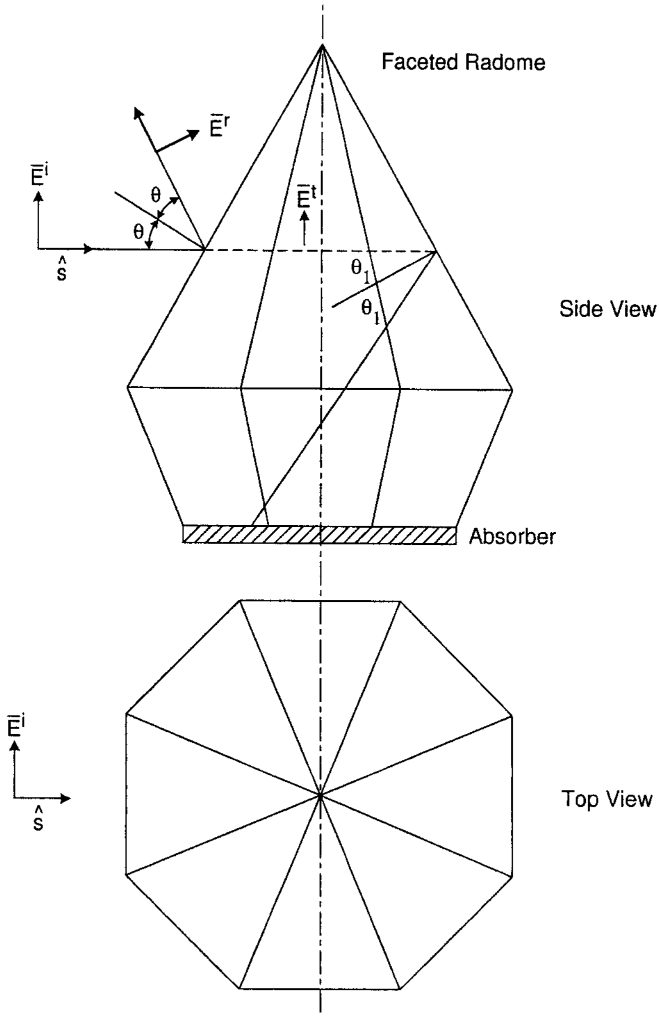


Fig. 4.25 A typical octagonal faceted radome. An incident signal in the horizontal plane will be reflected upward as shown if the radome is opaque. If part of the incident signal leaks through the front, it will essentially be reflected downward by the back. An absorber can be placed on the floor to prevent further scattering.

The situation becomes even more complicated if the frequency is somewhere in the transition range between being opaque and transparent. In that event, part of the incident signal may simply make it to the back wall where it will be partly reflected either down into the ground or upward. Although both of these signals will be reflected away from the horizontal plane, it is usually a good idea to cover the floor inside the radome with an absorber as indicated in Fig. 4.25.

We are at present unable to model a large faceted radome as shown in Fig. 4.25. However, we may combine eight *finite* × *infinite* flat panels into an octagon shape as indicated in Fig. 4.26. There is a fundamental difference between the finite radome in Fig. 4.25 and the infinite long one in Fig. 4.26—namely, that in the latter case, reflections from the back wall will occur and can in that case obscure the reflections from the front. This problem can in our calculations be alleviated by covering the back wall with an absorber or, simpler yet, by removing the four back panels as indicated in Fig. 4.26. Thus, in the cases to follow we shall simply calculate the backscattered field from just the four front panels as also indicated in Fig. 4.26.

We note that the angles of incidence are 22.5° for the two front panels, while they are 67.5° for the two side panels. Based on our observations above (see Section 4.13), we may conclude that we should use at least five and most likely more loaded columns on each side of the corner columns. Furthermore, we would anticipate that the loading should be somewhat less severe than earlier since we are now talking about "bending" of the panels rather than a complete interruption.

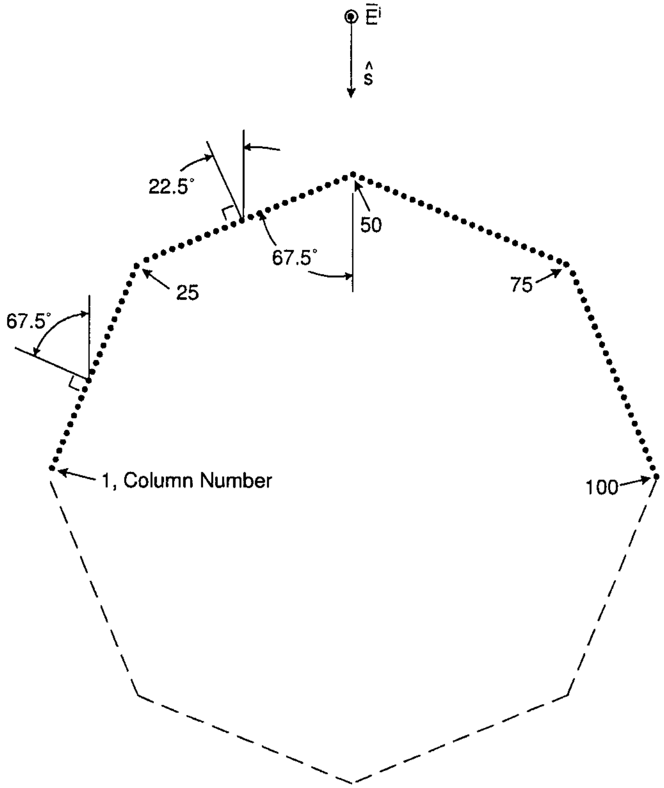


Fig. 4.26 Crude modeling of the faceted radome shown in Fig. 4.25. It is comprised of finite × infinite planar arrays. However, to avoid possible reflections from the back, only the four front panels are used to study the column currents in the front.

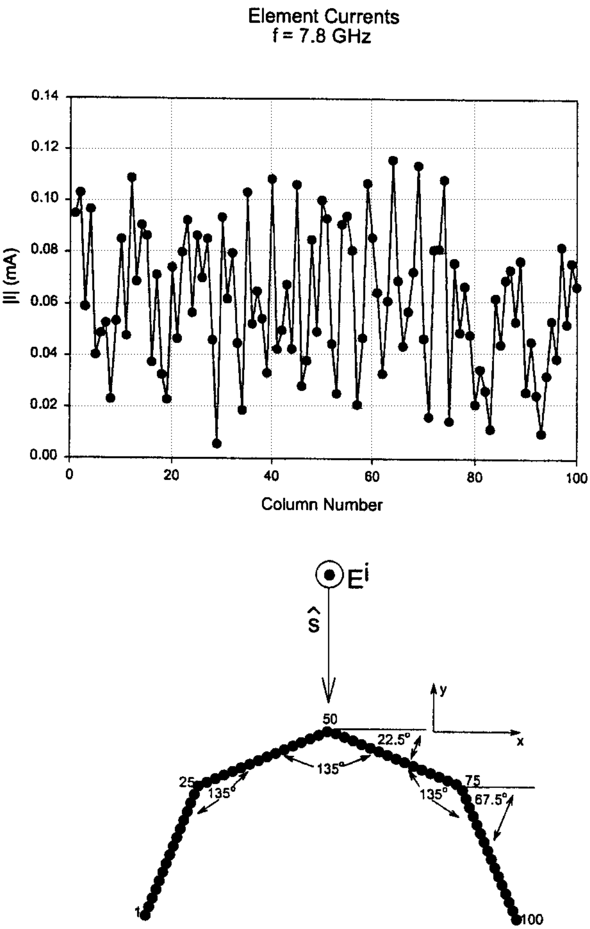


Fig. 4.27 The calculated column currents (from SPLAT) for the four front finite × infinite panels of an octagonal-shaped radome. Number of columns in each panel is 25, total number of four front panels is 100. There is no resistive loading; that is, this is the baseline case.

We shall present several practical examples obtained by use of the SPLAT program discussed earlier.

First we show in Fig. 4.27 the column currents at $f = 7.8$ GHz in the four front panels when no loads are present at all. This case constitutes our baseline. Note the very dramatic variation of the column currents as expected based on our investigation above (as discussed already in Chapter 1 and in this chapter as well, such a strong variation will only be expected in the frequency range where strong surface waves can exist, namely typically 6.3–8.5 GHz and not at other frequencies).

Next we show in Fig. 4.28 the same four front panels as shown above but now where five columns on each side of the corner columns have been resistively loaded as indicated in the figure (note the change of scale). We observe a very

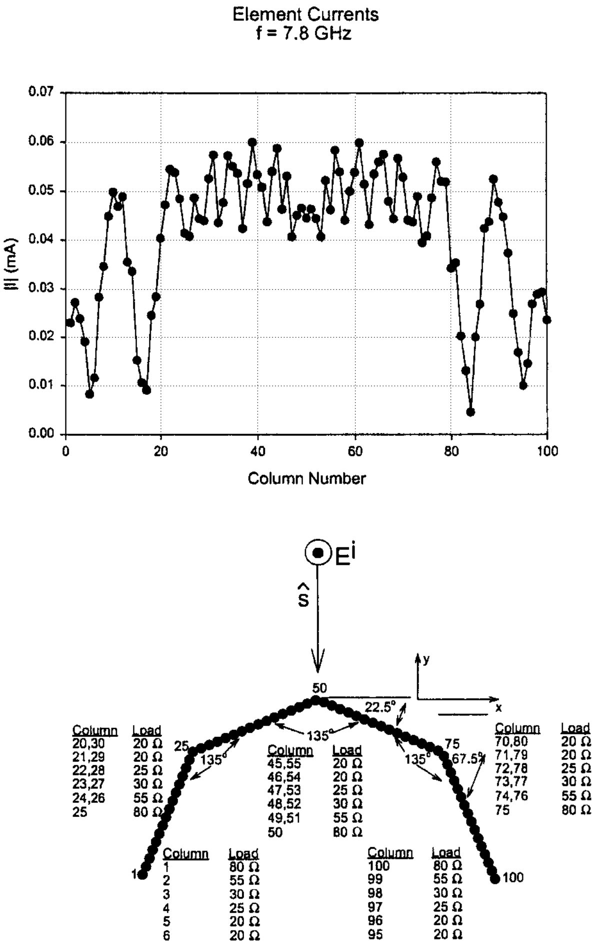


Fig. 4.28 The calculated column currents (from SPLAT) for the four front finite × infinite panels of an octagonal-shaped radome, like those shown in Fig. 4.27. However, the corner and end columns have been loaded resistively at the center of each element as indicated in the insert. Note the dramatic reduction in variation of the element currents as compared to the unloaded case in Fig. 4.27. This indicates much lower amplitudes of the surface waves and thereby lower scattering. (Note the different scales in Figs. 4.27 and 4.28.).

dramatic reduction of the column currents in the two front panels, while the variation in the two side panels still is significant but has been "cleaned up."

In Fig. 4.29 we show an interesting verification of some concepts developed earlier in this chapter. In Figs. 4.5 and 4.12 we described the surface waves as being excited by two imaginary semi-infinite arrays. We also demonstrated that this excitation could be reduced by insertion of barriers in the form of resistively loaded columns. Furthermore, we observed that the strongest surface wave would be excited at the corners at columns 25 and 75 and relatively weakly at columns

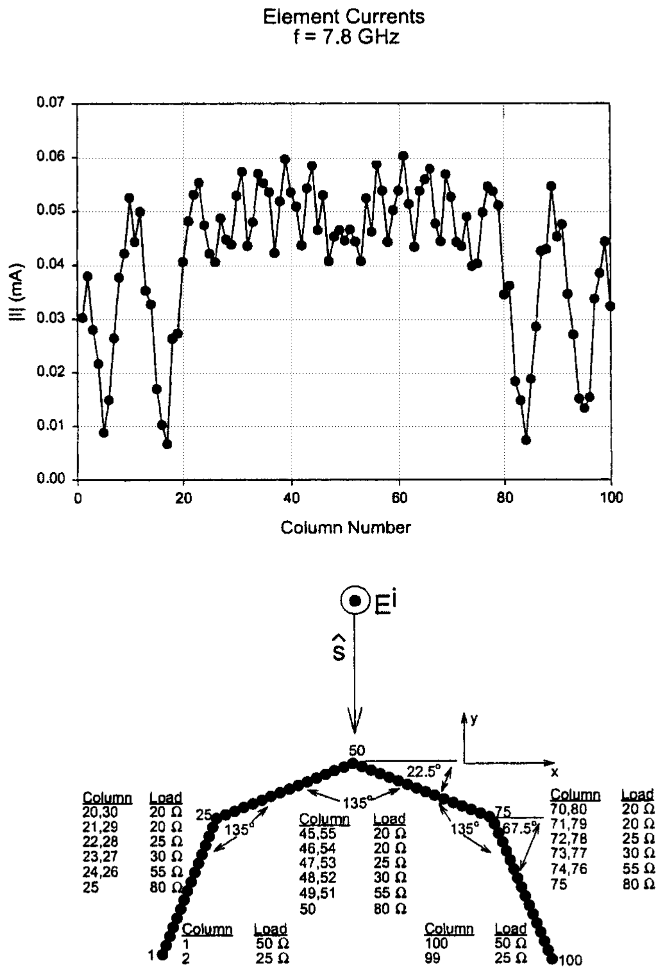


Fig. 4.29 The calculated column currents (from SPLAT) for the four front finite × infinite panels of an octagonal-shaped radome, like those shown in Figs. 4.27 and 4.28. However, only the two end columns have been resistively loaded as shown in the insert. Note the very little difference between the cases shown in Figs. 4.28 and 4.29.

1 and 100 (for illustration of this statement consult Fig. 4.5). Thus, the barrier at the open ends of the four panels need not be very wide. In fact, the main reason for having loaded columns at all in this case is simply to absorb the surface and Floquet waves incident upon the open edge.

Thus, we show in Fig. 4.29 a case where the number of loaded columns at the open ends has been reduced to just two columns. Sure enough, the two cases in Figs. 4.28 and 4.29 look very much alike except for the amplitudes of the column currents at the open ends. That could easily be adjusted by changing the load resistors at the open ends (if needed).

We may conclude from the examples above that surface waves become more prevalent at the higher angles of incidence. Also the interference wavelength between the Floquet and the surface waves becomes longer. This necessitates using more loaded columns at higher angles of incidence, which fortunately also leads to improved performance at the lower angles of incidence. Further good news is the fact that the backscatter is somewhat lower at the higher angles of incidence than at the lower ones because the sidelobe level of the scattering pattern is lower.

4.16 EFFECTS OF DISCONTINUITIES IN THE PANELS

Small faceted radomes can be made by flat panels containing no discontinuities. Larger ones will in general be constructed of large flat panels comprised of smaller panels. A pertinent question then becomes how precise must these small panels be joined together in order to avoid excitation of surface waves.

Considering the complexity of the typical periodic surfaces used today, we must state that a precise answer to this question is not possible, nor do we need it. Here we shall limit ourselves to investigating the effect of introducing discontinuities somewhere in the flat panels.

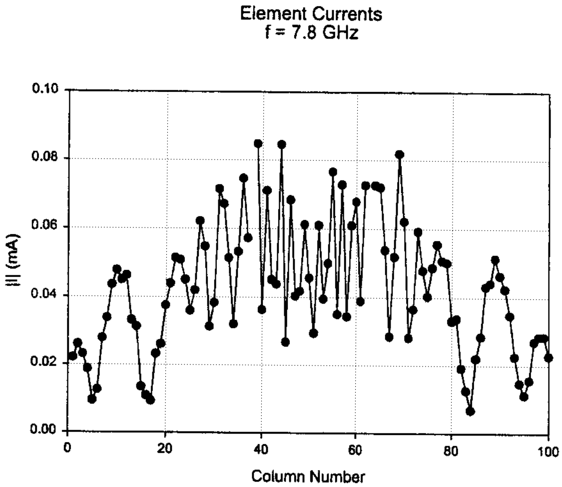
More specifically we show in Fig. 4.30 the case shown earlier in Fig. 4.28 but where a discontinuity has been introduced in each of the two front panels by simply removing one of the unloaded columns located in the middle of the two front panels. Similarly we show in Fig. 4.31 the case where two unloaded columns have been removed from each of the side panels. We note a considerable effect when the discontinuity is at the two front panels and very minor when it is at the sides. Part of the explanation for this observation might be that the surface waves are considerably stronger in the side panels already.

Based on our observations earlier in this chapter, we may conclude that the amplitudes of the surface waves in these examples are sufficiently low not to warrant any significant radiation and thereby raise the RCS level. Unfortunately, precise numbers of the RCS level cannot be reported.

4.17 SCANNING IN THE *E* PLANE

So far in this chapter we have considered plane waves incident in the *H* plane only. In general this is the plane where the most important phenomena occurs. However, one should be cognizant of the fact that surface waves characteristic for finite structures can also exist for *E*-plane incidence. We shall explore this case in some detail in the following.

In order to make meaningful comparisons between the two cases, we shall consider an array with the same element length $2l = 1.5\,\text{cm}$ as in the *H*-plane case above. However, the array will be slightly modified as explained in Fig. 4.32. First we show in Fig. 4.32a the original array used for the *H*-plane case. If



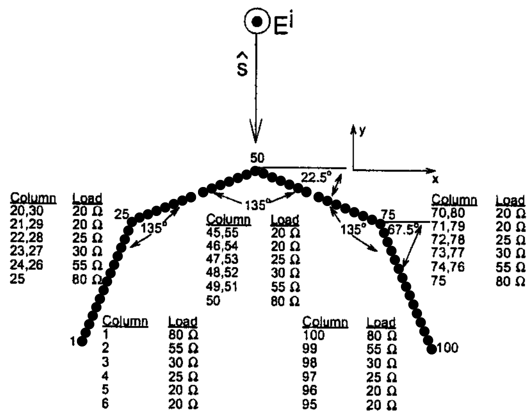


Fig. 4.30 The effect of a discontinuity in each of the two front panels of radome in Fig. 4.28 by simply removing a complete column in the middle of each one as indicated in the figure. Note change of scale.

we would scan this array in the E plane, we would readily observe that the first grating lobe for grazing incidence would occur at $\lambda_{G.L} = 2D_z = 3.2\,\text{cm}$ or $f_{G.L.} = 9.38\,\text{GHz}$. Considering that the resonance frequency is around 10 GHz, this simply is too low even if surface waves typically occur at frequencies somewhat below resonance (see Sections 4.7 and 4.8).

However, early onset of grating lobes can in this case be easily prevented by interlacing the elements as shown in Fig. 4.32b. Note that array dimensions remain the same in the two cases except that adjacent columns have been shifted with respect to each other as shown.

Finally we recall that the SPLAT program is structured to have the finite dimension along the x axis and the infinite along the z axis. Thus, the array

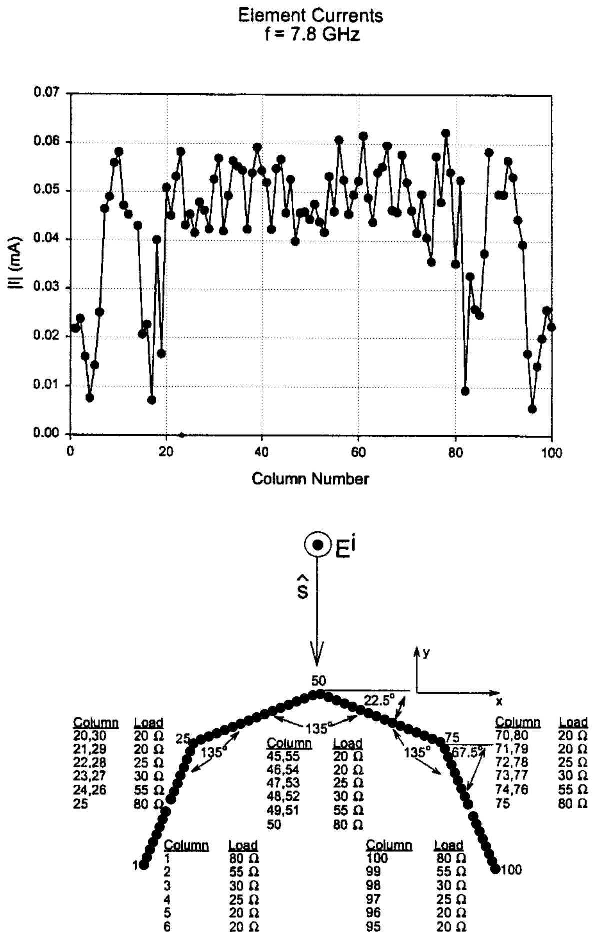


Fig. 4.31 The effect of a discontinuity in each of the two side panels of the radome in Fig. 4.28 by simply removing a complete column in the middle of each one as indicated in the figure.

in Fig. 4.32b must be rotated 90° as shown in Fig. 4.32c. Note that all array dimensions are the same in the two cases. Only the x and z axes have been interchanged. The dimensions in Fig. 4.32c will be used in the following.

We shall investigate the E-plane case analogous to the H-plane case above, namely by plotting the scan impedance from broadside ($s_x = 0$) and all the way into the "end" of imaginary space and back into real space.

A typical example of an E-plane scan impedance for an infinite array is shown in Fig. 4.33. We start at broadside at $\eta = 0°$ for $s_x = 0$ and proceed to grazing at $\eta = 90°$ for $s_x = 1$, which marks our entrance into the imaginary space. As s_x becomes larger than 1, we see that the scan impedance moves downward along the imaginary axis until it reaches its lowest level for $s_x = 1.12$ from where it starts moving upward and eventually crosses the real axis for $s_x = 1.65$. For

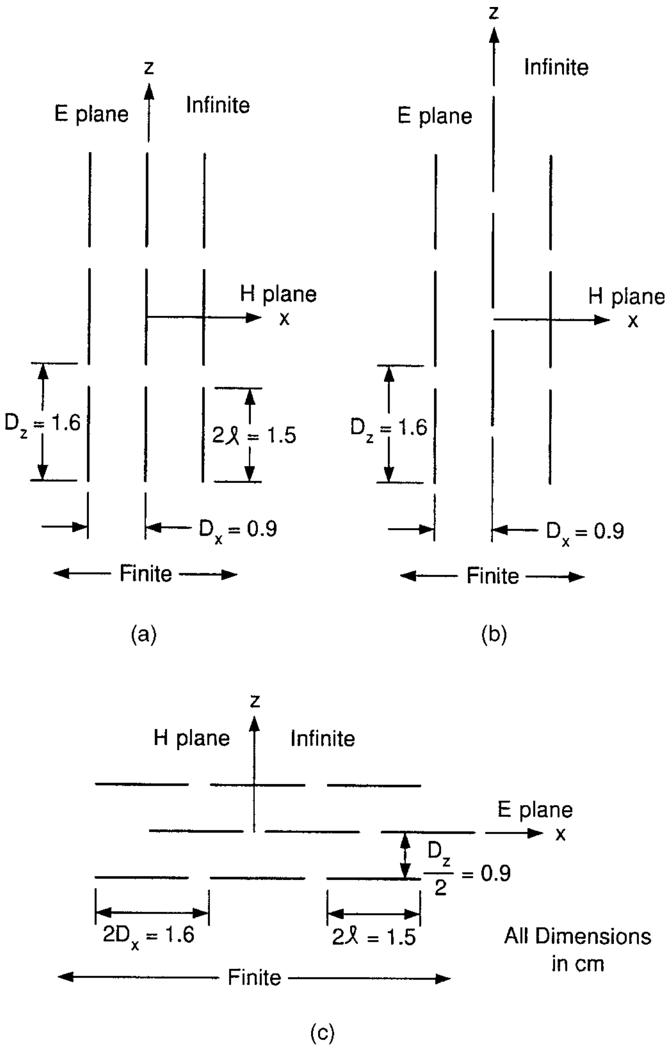


Fig. 4.32 Modification of the array when changing from H-plane to E-plane scan. (a) The original array used for H-plane scan. If we scan this array in the E plane, grating lobes will start too early, namely at 9.38 GHz. (b) By interlacing adjacent columns as shown, the onset of grating lobes can be delayed to a much higher frequency. (c) To comply with the structure of the SPLAT program the array in (b) is rotated 90° and the x and z axis interchanged as explained in the text.

$s_x = 2.08$ we reach the upper limit of the scan impedance. Increasing s_x beyond 2.08 produces the identical scan impedance on the way out of the imaginary space as on the way in until we reach $s_x = 2 \times 2.08 = 4.16$ or 0. The scan from $s_x = 4.16 - 1$ to 4.16 corresponds to $s_x = -1$ and 0; that is, the scan goes from $\eta = -90$ to $0°$ (see also Fig. 4.2).

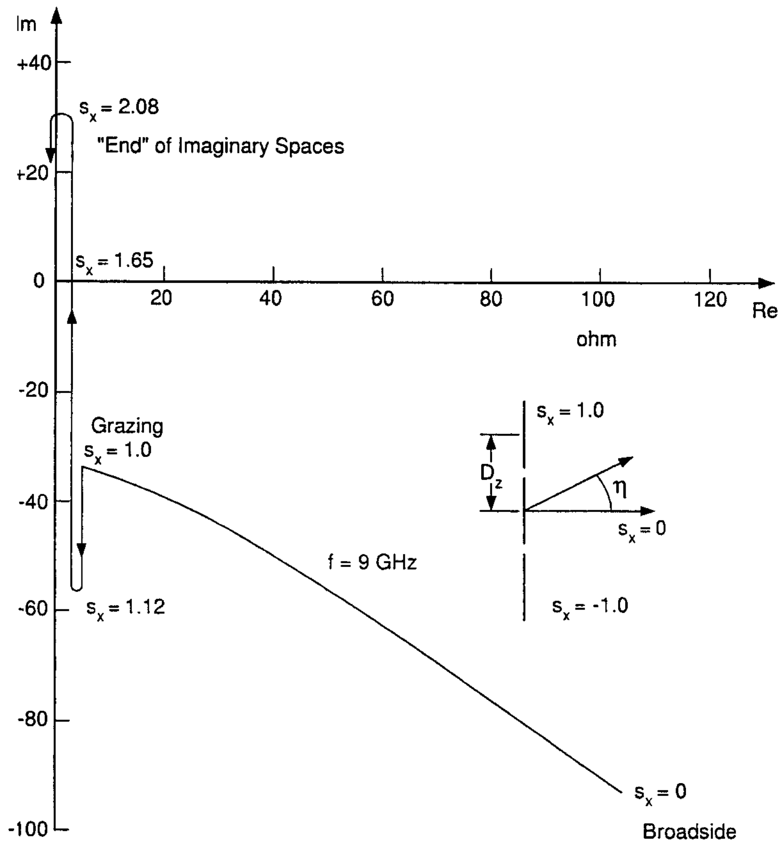


Fig. 4.33 The scan impedance at $f = 9\,\text{GHz}$ when the array in Fig. 4.32c is scanned in the E plane. Broadside starts at $s_x = 0$ and grazing is at $s_x = 1.0$. Higher values of s_x correspond to scan in the imaginary space where the scan impedance is purely imaginary. It gets to the "end" of imaginary space for $s_x = 2.08$ from where it goes right back down the way it got in.

The curve shown in Fig. 4.33 is an adaptation of data obtained from Janning's dissertation [77]. He used three current segments along each element since an asymmetric current distribution could be expected for E-plane scan. However, he also notes that good results could be obtained by using just a single current segment since the length of the elements in the present case is somewhat shorter than one-half λ.

There is a significant difference between the H-plane scan in Fig. 4.2 and the E-plane scan in Fig. 4.33—namely, that the former may only pass close to the origin for one value of s_x while the latter potentially can pass close to the origin for two values of s_x. At higher frequencies the impedance curves will move upward and vice versa at lower frequencies as shown specifically in Fig. 4.34 for the frequencies $f = 8\,\text{GHz}$ and $10\,\text{GHz}$. Obviously no strong surface wave can exist at either of these two frequencies, but at frequencies somewhere in between

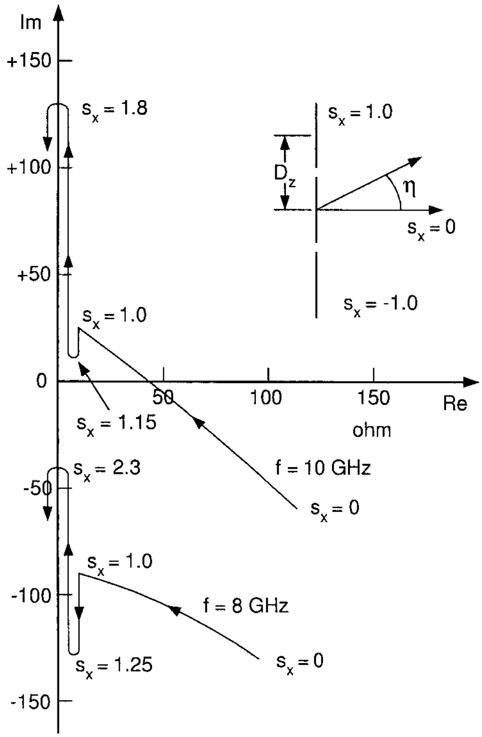


Fig. 4.34 Same scan impedances as shown in Fig. 4.33 but at f = 8 and 10 GHz, respectively. Note how the higher frequencies move up toward the inductive region of the complex plane and vice versa for the lower frequencies. At some frequencies between 8 and 10 GHz we have the possibility to pass close to the origin once and even twice.

we may pass close to the origin for one and even two values of s_x. That simply means that instead of one left- and one right-going surface wave we may have two left- and two right-going surface waves.

An example of actual calculated double surface waves are shown in Fig. 4.35. We clearly observe the Floquet current at $s_x = -0.707$ while we also observe a pair of right-going surface waves at $s_x = 1.015$ and 1.165 as well as left-going surface waves at $s_x = -1.015$ and -1.165. It is noteworthy that the column currents in Fig. 4.35 were produced by a simple Fourier transform of the actual calculated currents obtained directly from the SPLAT program. Thus, their validity can hardly be in doubt.[3] Therefore, even if one does not accept the physical explanation presented above, the facts speak fairly loudly. There will, however, always be those who refuse to believe anything they do not readily understand because things are explained somewhat differently than they are used to. And that is of course their right.

[3] The double surface waves were observed first by Janning (see his dissertation [77]) and are therefore often referred to as "Janning's Anomaly."

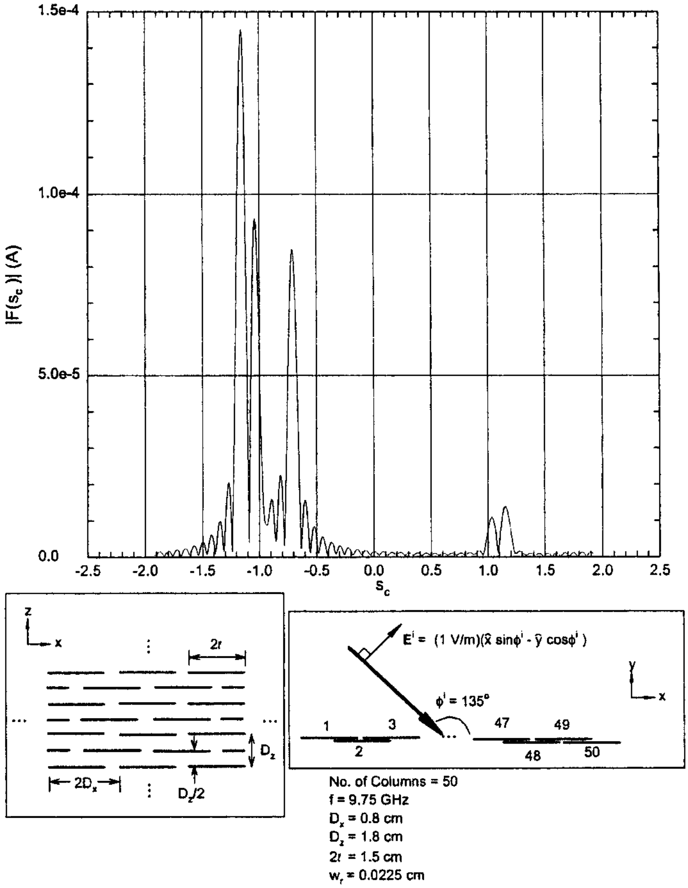


Fig. 4.35 The column currents at $f = 9.75$ GHz for E-plane scan showing the Floquet currents at $s_x = -0.707$ and double surface waves at $s_x = \pm 1.015$ and $s_x = \pm 1.165$. The double surface waves were observed first by Janning (see his dissertation [77]) and are therefore often referred to as "Janning's Anomaly.".

But I would like to have these individuals explain these phenomena "their way."

4.18 EFFECT OF A GROUNDPLANE

So far we have considered surface waves only on finite periodic structures without a groundplane. When a groundplane is added to an array of dipoles, it is usually driven actively. This case is in practice somewhat different from the passive case considered above by the fact that all elements are connected to generators or amplifiers with impedances comparable to the scan impedances. As explained in Chapter 5, this leads to a highly desirable attenuation of any potential surface waves.

There is, however, a case where load resistors at each element are just not available, for example, when a slotted periodic surface is placed next to another slotted surface as is the case in a biplanar radome. In that event, surface waves can be excited on the finite slotted surface facing the incident signal while the second periodic surface essentially acts as a groundplane (at least in the frequency range where surface waves can potentially exist). Obviously the outer slotted surface does not enjoy the benefit of resistive loads connected to each slot.

The question then becomes, How will the second slotted surface affect the frequency range of the surface wave on the first?

From a physical point of view we note that the field associated with the surface wave on the latter is composed of an assembly of evanescent waves that is attenuated as we move away from the periodic structure. In fact these evanescent waves have fallen to rather low values at a distance of a quarter wavelength. Since this is about the typical spacing between adjacent slotted surfaces in many applications, we may conclude that the surface wave frequency for a single periodic surface is only lightly affected by the presence of a groundplane or a second slotted surface.

Janning examined in his dissertation [77] the effect of a groundplane upon arrays of dipoles. His approach was completely rigorous (actually he wrote a very fine dissertation that is highly recommended). His findings essentially verify the physical arguments put forth above.

4.19 COMMON MISCONCEPTIONS CONCERNING ELEMENT CURRENTS ON FINITE ARRAYS

4.19.1 On Element Currents on Finite Arrays

In Sections 4.6 and 4.7 we explained how the total currents of a finite array could be decomposed. Although the concept actually was quite simple, it often leads to some misconceptions. Some of these will be illustrated in Fig. 4.36 and discussed in the following.

In Fig. 4.36a we show the currents on an infinite array being exposed to an incident plane wave. As is well known, it results in currents of equal amplitude and phases matching that of the incident plane wave, namely based on the so-called Floquet Theorem. We next show in Fig. 4.36b how we can obtain a finite array, by superimposing two semi-infinite arrays with negative Floquet currents on the infinite array case in Fig. 4.36a. Obviously, that leads to zero current outside the finite array (exact). However, the finite array will itself have currents comprised of the original Floquet currents caused only by the incident plane wave plus some unknown currents caused only by the fields from the two semi-infinite arrays with negative Floquet currents. Note that the currents on the two semi-infinite arrays are defined very precisely as simply being equal to the negative of the original Floquet currents. This by no means implies that the currents on two semi-infinite arrays by themselves are simply of the Floquet

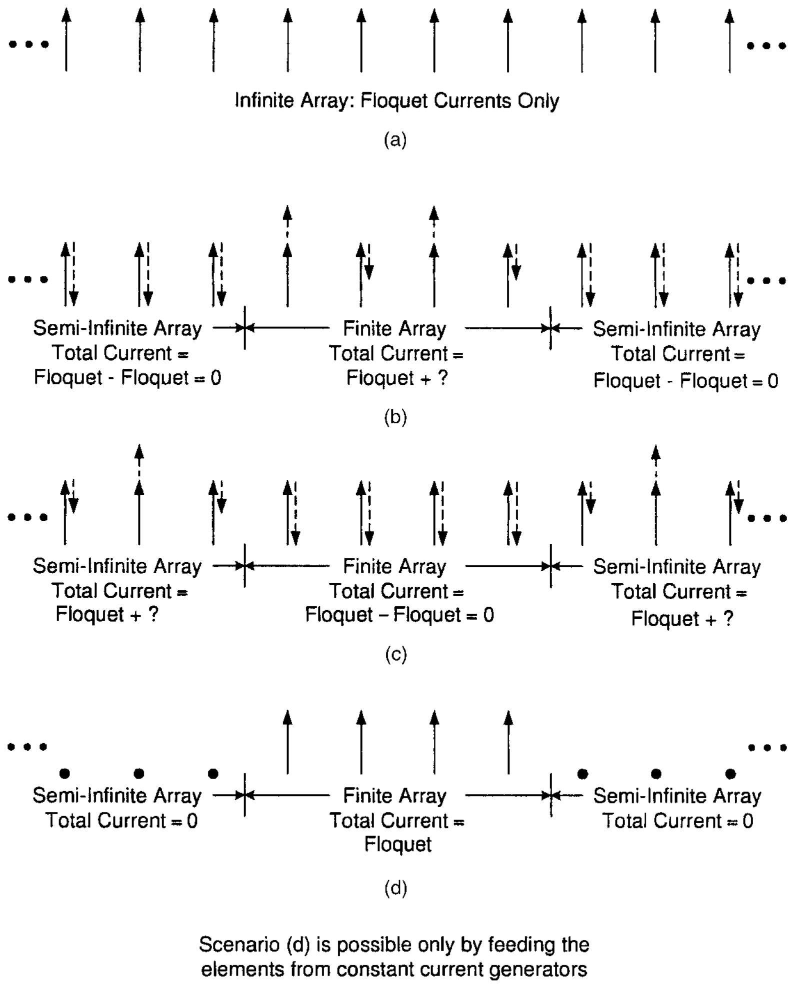


Fig. 4.36 Some interesting finite and semi-infinite array configurations. (a) The amplitudes of the column currents on an infinite array being exposed to an incident plane wave. Consists of Floquet currents only. (b) Creating a finite array is done by adding two semi-infinite arrays with negative Floquet currents (exact) outside the finite array. The new currents in the finite array are complicated (see text). (c) Alternatively, the currents on two semi-infinite arrays can be obtained by superimposing a finite array with negative Floquet currents only upon the infinite array in (a). The new currents in the semi-infinite arrays are complicated (see text). (d) A finite array with Floquet currents is possible only when feeding the elements from constant current generators. An incident plane wave induces a voltage as a constant voltage generator and can therefore not produce Floquet currents only. The currents are then determined as in case (c).

type when exposed to an incident plane wave. In fact, this case is illustrated in Fig. 4.36c. Here a finite array with negative Floquet currents are superimposed on the original infinite array, case Fig. 4.36a. There are now no currents between the two semi-infinite arrays, while the currents on these consist of the original

Floquet current caused by the incident plane wave plus currents induced by the fields from the finite array between them and with negative currents.

Note again that the currents on neither the finite arrays in Fig. 4.36b nor the two semi-infinite arrays in Fig. 4.36c will in general ever consist of merely the Floquet currents. In fact, a finite array with only Floquet currents is shown in Fig. 4.36d. The only way a finite array can have constant element currents is by feeding them from constant current generators. While this is possible when feeding a finite structure as a phased array from man-made generators, an incident plane wave will merely induce a voltage in each element and not a constant current.[4] The fact that many researchers assume constant currents in finite arrays when exposed to an incident plane wave is simply an approximation that leads to erroneous conclusions. These cannot be remedied by application of more or less elaborate mathematical manipulations like, for example, Poisson's sum formula adapted to semi-infinite arrays with constant current. The starting point is simply too gross an approximation to obtain meaningful insight into a much more complex problem.

We finally remind the reader that the discussion above merely serves the purpose of explaining the mechanism of finite arrays. The actual decomposition into Floquet and surface wave currents as well as end currents is done by applying a Fourier analysis of the actual calculated currents obtained from the SPLAT program. Thus, whether one accepts the explanation above or not, the results speak for themselves.

4.19.2 On Surface Waves on Infinite Versus Finite Arrays

In this chapter much attention has been focused on surface waves that can exist only on finite arrays. The reactions to these findings are often sharply divided. Some deny that they actually exist. Other feel intuitively: "Yes, of course they can exist. But they will be there even on an infinite array, so why make all that fuss out of finite arrays?"

Although an extensive discussion of this subject was given in Section 4.6, it is nevertheless a fact that many readers do not quite understand our argument or simply do not have sufficient time to digest it. They would much prefer the following four-sentence contradiction.

Let us assume that a surface wave with a phase velocity depending only on the infinite structure was indeed present. Since the phase velocity of the Floquet currents depend only on the angle of incidence, the two waves would produce an interference wave with a wavelength unrelated to the periodicity of the array as shown, for example, in Fig. 4.19a. That would violate Floquet's Theorem, which is valid for an infinite array only, not a finite. Thus, the new kind of surface wave can exist only on the finite array.

[4] Only in the infinite array case is it immaterial how the elements are fed. Whatever the type of generator, we will always obtain merely Floquet currents.

4.19.3 What! Radiation from Surface Waves?

In our EM upbringing we have been inundated with numerous "facts" of life. One of these is, Surface waves do not radiate!

We can indeed agree with this "theorem," but only if the surface wave is attached to an infinite, straight structure. This point is stated very clearly in the discussion associated with Fig. 4.2. However, as also indicated in the same figure, we do indeed experience radiation when the periodic structure is finite. This phenomena is often interpreted as radiation emanating exclusively from the ends of the periodic structure while nothing comes from the rest of the structure. This is an observation, not an explanation. Even if it appears to come from the extremities of the structure, we should never lose sight of the fact that EM fields originate from electric and/or magnetic ($\overline{m} = \overline{E} \times \overline{n}$) currents. Thus, the total radiation from a finite periodic structure is obtained by integration of the currents present on the *entire* structure completely analogous to a classical antenna problem. See Section 4.8 for details.

However, once the total radiation is determined, it is of course a simple matter to assume that the radiation emanates entirely from the ends of the structure and then obtain some magic coefficients that potentially could be labor-saving, was it not for the fact that these are not invariant but change considerably with frequency and angle of incidence (see Section 4.9).

Thus, these "magic coefficients" should be traded among scientists with great reservation. And not like small boys trade baseball cards!

4.20 CONCLUSION

Much of the spotlight in this chapter has been focused upon surface waves on passive periodic structures. It appears that at this point in time we have two distinct groups, one of which is associated with the presence of a stratified medium placed in the immediate neighborhood of the periodic structure. It always requires a stratified medium to exist but is independent upon whether the structure is finite or infinite. It readily shows up in programs based on the infinite array approach like, for example, the PMM program.

The other group can exist whether a dielectric is presented or not, but the structure must be finite. Thus, it shows up only in programs based on finite array theory like, for example, the SPLAT program and not when using the PMM program.

The first type (Type I) can be viewed merely as grating lobes trapped inside the stratified dielectric medium. Therefore they typically occur at frequencies above resonance, namely such that grating lobes has started to emerge inside the stratified medium.

In contrast the second type usually occurs only in a band of frequencies considerably below the resonance frequency, like 20–30% for H-plane scan and 10% for E-plane scan and only when the interelement spacing is less than $\lambda/2$.

Both types can radiate and thereby lead to enhanced scattering. They are consequently in general not very desirable.

The first type can usually be avoided by simply making the interelement spacings sufficiently small in terms of wavelength. This approach usually leads to periodic structures of superior stability of the resonance frequency with angle of incidence. Thus, there is no conflict here.

The second type (Type II) is usually controlled by resistively loading the elements in one or more columns located at the edge of the finite structure. It is also possible to lightly load all elements in the entire periodic structure; however, this approach leads to reflection and transmission loss and is therefore in general not recommended (however, when dealing with active surfaces rather than passive ones, it is OK; see Chapter 5).

An important application of finite flat panels is for making faceted radomes. They must be treated at the edges resistively in order to attenuate the surface waves. Otherwise the RCS of such radomes will be larger than expected (in fact, this was how the presence of surface waves was first suspected).

Large faceted radomes are often made by joining smaller flat sections into larger flat sections. We demonstrated that the grooves between panels must be done very carefully in order not to create additional surface waves.

If the radome is curved rather than flat, there will be a natural attenuation due to shedding of energy along the surface. Since many radomes and dichroic surfaces in use today are in fact curved, this is probably the reason that this new type of surface wave has gone largely unnoticed.

PROBLEMS

4.1 The elements in Fig. 4.16 are encapsulated in dielectric tubing with wall thickness equal to the wire diameter and relative dielectric constant equal to 3. Without actually solving the problem, estimate and explain what will happen to the impedance curves in Fig. 4.16.

How will the frequency of the surface waves change?

This encapsulation can actually be done in the SPLAT program as written by Usoff [24].

4.2 Instead of encapsulating the elements in dielectric tubing, embed the elements in a dielectric slab (finite, please!) with a total slab thickness equal to the diameter of the dielectric tubes.

How will the new surface wave frequency compare with the values obtained in Problem 4.1?

To the best of the author's knowledge, this problem has not been solved rigorously for the finite dielectric slab.

If anyone out there pulls it off, I definitely would like to hear about it!

4.3 In this chapter we saw several examples where a finite periodic structure exposed to an incident plane wave of oblique angle of incidence could, due to surface waves, exhibit a backscatter several decibels higher than expected when based on simple Floquet currents only.

By inspecting several calculated bistatic scattering patterns in this chapter, discuss the loss of the reflection coefficient in the bistatic direction. Is this problem serious?

Similarly, if we instead consider an array of slots, discuss whether the loss in transmission coefficient in the forward direction would be of great concern.